



Design and Development of an AI-Powered Universal Smart Remote for Seamless Control of Home Appliance



A MINI PROJECT II REPORT

Submitted by

HARIPRASATH P (71812302045)

JOVIKESH P M (71812302053)

KATHIR N (71812302057)

In partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

in

DEPARTMENT OF ELECTRONICS AND COMMUNICATION

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

ANNA UNIVERSITY : CHENNAI 600025

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SRI RAMAKRISHNA
ENGINEERING COLLEGE



[Educational Service : SNR Sons Charitable Trust]

Vattamalaipalayam, N.G.G.O. Colony Post,
Coimbatore – 641022.

 www.srec.ac.in

 +917530089996

ANNA UNIVERSITY: CHENNAI 600025

BONAFIDE CERTIFICATE

Certified that this project report entitled “ **Design and Development of an AI-Powered Universal Smart Remote for Seamless Control of Home Appliances**” is the bonafide work of **HARIPRASATH P (71812302045), JOVIKESH P M (71812302053), KATHIR N (71812302057)**, who carried out the **20EC280** Mini Project II under my supervision.

SIGNATURE

SUPERVISOR

Mrs. Shanthini E

Assistant Professor, (SI.G)

Department of ECE

Sri Ramakrishna Engineering College

College Coimbatore - 641022

SIGNATURE

HEAD OF THE DEPARTMENT

Dr. Jagadeeswari M,

Professor and Head

Department of ECE

Sri Ramakrishna Engineering

Coimbatore – 641022

Submitted for Project Viva Voce Examination held on _____

Internal Examiner

External Examiner

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ABSTRACT

A novel idea for the design of an intelligent and compact universal smart remote system that can be used to control several household and classroom appliances using a single device is proposed. The intelligent universal remote system utilizes the ESP32 microcontroller and makes use of advanced technologies like gesture recognition and machine learning. This paper proposes the use of an MPU6050 sensor for hand movement detection and classifies data using decision tree and K-nearest neighbors (KNN) algorithms for real-time and precise gesture recognition. An infrared (IR) receiver is used in the system for receiving signals from traditional remote controls, and an IR transmitter is used for sending the captured signal to control appliances, hence making it compatible with existing devices without modifications. The gesture-based interface does not have any physical buttons and operates in a touch-free manner. Although the performance of the system initially using only IR is functional, the use of machine learning in the process greatly improves the system's performance. The proposed system achieves high gesture classification accuracy with low latency and efficient memory utilization suitable for embedded implementation. It demonstrates reliable real-time performance with high operational efficiency and stable control execution. The compact design, combined with multi-mode control capability and intelligent automation, makes the system suitable for applications in smart homes, classrooms, and assistive environments. Furthermore, the system can be extended with IoT connectivity and voice control features for future smart automation. The proposed solution offers a cost-effective, scalable, and energy-efficient alternative to traditional remote control systems, enhancing user convenience and enabling personalized appliance management.

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LIST OF ABBREVIATIONS

ESP32	Espressif 32-bit Microcontroller
MPU6050	Motion Processing Unit (Accelerometer + Gyroscope)
IR	Infrared
ML	Machine Learning
KNN	K-Nearest Neighbors
DT	Decision Tree
IoT	Internet of Things
OLED	Organic Light Emitting Diode
I2C	Inter-Integrated Circuit
GPIO	General Purpose Input Output
AI	Artificial Intelligence
MCU	Microcontroller Unit
PWM	Pulse Width Modulation
UART	Universal Asynchronous Receiver Transmitter
ADC	Analog to Digital Converter

CHAPTER 1

INTRODUCTION

1.1 Background of the Study:

With the development and growth of smart home technology and embedded systems, there is a growing need for devices that can offer greater convenience and easier control of electronic devices. In our homes, classrooms, and offices, there are many electronic devices used for entertainment and presentation purposes. These include televisions, air conditioners, projectors, and set-top boxes. These electronic devices are normally controlled by individual infrared remotes. Controlling these electronic devices using individual infrared remotes has presented many challenges. These challenges include users losing remotes, difficulties in memorizing functions for individual remotes, and overall inconvenience in controlling these remotes. This has resulted in inefficiency in controlling these remotes for our day-to-day activities. In order to overcome these challenges and limitations in controlling these remotes, there is a need for a universal remote control system. This will be able to replace all remotes for individual electronic devices with a single remote control device. This will be achieved by incorporating advanced embedded technology into universal remotes. This will make these remotes more efficient and convenient for users. The proposed system aims at developing a smart universal remote control device for controlling infrared remotes for individual electronic devices.

1.2 Requirement of Universal Remote Control Systems:

These days, pretty much every electronic thing at home uses those infrared remotes to work. Like TVs, air conditioners, sound systems, all of them come with their own little controller. So if you have a bunch of devices, you end up with this pile of remotes scattered around. It gets

annoying fast. I mean, trying to keep track of what one does, and sometimes you fumble around looking for the right one when you just want to turn something on quick. That complexity makes everything harder, and yeah, losing one or two is super common. A universal remote could fix that, I think. It basically combines all the functions from different remotes into just one device. No more juggling a handful of them. Users interact with their stuff way simpler, without all the hassle. It seems like that would cut down on the problems a lot, though maybe not everything is perfect about it.

Using gestures to control things gets rid of those physical buttons, which really helps make the whole system easier to use. It feels more natural that way, kind of like how we move our hands without thinking about it. This part about intuitiveness stands out, I mean, it just flows better for people. So, yeah, we need something that puts infrared signals together with recognizing those gestures. That should give a solid, up to date way to handle controls, even if its not perfect yet.

1.3 Importance of Efficient Signal Processing in IR Systems:

Infrared communication basically works by sending and getting these encoded signals from a remote to whatever device you're controlling. I think the whole thing hinges on how accurate that transmission is, because if its off even a little, the system just doesn't function right. Signals from all sorts of remotes come in different formats, you know, with varying timings and stuff. So, to make it work, you have to decode them properly, store them away, and then send them back out when you need to control something. It seems like any kind of mismatch in processing those signals could really screw up the device, make it act wrong or not respond at all. For handling a bunch of these IR codes, the system needs to process them efficiently, storing multiples and pulling them up accurately later. Timing and modulation have to be spot on too, especially when the signal goes through the IR LED for transmission.

Implementing good techniques for signal handling lets the system talk reliably to different devices. Overall performance gets better, and the universal remote operates more consistently. Some people might think its straightforward, but errors can still sneak in there.

1.4 Introduction to the Proposed System:

The proposed system is a gesture-controlled universal remote controller implemented with an ESP32 microcontroller. This system will utilize infrared communication and motion sensors for a smart control interface. The proposed system will have the following major components: ESP32 microcontroller, MPU6050 accelerometer and gyroscope sensor, infrared receiver module, and infrared transmitter module (LED). The infrared receiver module will be used for receiving infrared signals from existing remote-control devices, while the infrared transmitter module will be used for sending infrared signals to the devices.

The MPU6050 sensor will detect the gestures made by the user's hand. These gestures will be interpreted by the ESP32 microcontroller and will correspond to a set of commands. Depending on the gesture made, the system will send the corresponding infrared signal from its memory for controlling the device.

1.5 Autonomous Operation and System Features:

The proposed universal remote control system works on the basis of user gestures and stored infrared signals. The system will continuously read the motion data from the MPU6050 sensor and process the information to identify the predefined user gestures. Once the system identifies the user's gesture, the ESP32 microcontroller will execute the command by sending the stored infrared signal. This system does not require the user to continuously input

information using the buttons. Hence, it makes the system more user-friendly. The proposed system will be efficient and effective. It will perform the desired tasks only when the system identifies the user's gestures. This will prevent the system from continuously processing information. The system will also be more reliable because it uses embedded components. The proposed system will also be compact and portable. Hence, the system will be easy to use in different places such as homes, classrooms, and offices.

1.6 Advantages of the Proposed System:

The proposed system for a universal remote control has various advantages compared to the conventional method of controlling electronic devices. For instance, the proposed system eliminates the need for different controllers since it has the capability of controlling various devices with a single device. The proposed system uses gestures for controlling electronic devices. This eliminates the need for physical buttons, thus making it easy for users to interact with the devices. It uses infrared signals, which are commonly used in controlling electronic devices. The proposed system is cost-effective since it uses commonly available devices like ESP32 and MPU6050. It is also compact and easy to use, thus suitable for real-time applications. The proposed system is a suitable solution for controlling various electronic devices efficiently.

CHAPTER 2

LITERATURE SURVEY

Modern households and classrooms struggle with multiple IR remotes for TVs, ACs, projectors, and fans, causing clutter and line-of-sight issues. This project develops an AI-powered universal smart remote using ESP32, combining IR control, voice, gestures (MPU6050), and IoT for seamless operation. It learns user habits for personalized automation, improving convenience and efficiency.

2.1 INTRODUCTION

Current households and classrooms face significant challenges with appliance control. Multiple separate IR remotes are needed for devices like TVs, ACs, projectors, and fans, leading to clutter and confusion. Traditional remotes rely solely on line-of-sight signals, lacking smart features, while advanced systems limit to single modes like gesture-only or voice-only, without full integration or personalization. No affordable, unified solution exists for combining IR replay, AI learning, and IoT in everyday use. The proposed AI-powered universal smart remote overcomes these issues effectively. It unifies control of multiple appliances through IR, voice commands, gesture recognition, and IoT connectivity in one cost-effective device. By learning user habits, it enables personalized automation—like auto-activating AC at routine times—boosting convenience and energy savings in homes and classrooms. Scalable for future expansions, it reduces remote clutter, enhances accessibility, and supports seamless multi-modal operation without line-of-sight limitations.

2.2 BACKGROUND

Sam Young , Hao Zhou and Gursel Alici (2024) ‘Gesture Recognition Framework for Teleoperation of Infrared (IR) Consumer Devices Using a Novel pFMG Soft Armbani’, Sensors (MDPI), Vol.24, No.9, p.4123.

This project is about building a system that allows people to control devices like TVs, air conditioners, and other appliances that use infrared (IR) signals through hand movements. Instead of using traditional remote controls, the system uses a soft wearable armband that detects muscle activity in the forearm. This technology is called pFMG (pressure-based force myography), which measures how muscles expand and contract when performing movements. The goal is to create a natural way for people to interact with machines without the need for buttons or touchscreens. The main objective of the project is to provide an affordable and easy-to-use alternative to existing control systems. Many current gesture recognition methods rely on cameras, which can be unreliable. For example, poor lighting conditions or objects blocking the camera can affect accuracy. This system avoids those issues by using sensors that directly capture muscle movements. As a result, it works consistently regardless of external conditions like lighting. The system consists of multiple stages. First, the pFMG armband collects data from the forearm muscles when the user performs movements such as a fist or a wave. This raw data is then processed to remove noise and improve signal quality. Next, important features are extracted from the data to help distinguish one movement from another. These features are then fed into a machine learning model that classifies the movement in real time. Once the movement is recognized, it is mapped to a specific command. For example, one movement might turn the

Wenlong Zhang and Zhiwei Wang. (2022) ‘Incremental Online Learning Algorithms Comparison for Gesture & Visual Smart Sensors’, arXiv preprint arXiv:2207.06789.

This paper is about comparing different incremental online learning algorithms for stuff like gesture recognition and those visual smart sensor systems. I think the main point is to fix problems with old machine learning models that have to start over completely whenever new data shows up. In real life, like with gestures, things change all the time because of how users act or the environment, so the system needs to keep up without much hassle. The authors suggest using these incremental techniques where the model updates itself right away as new info comes in, and it does not forget what it already knows. That seems way better for cutting down on computer power and time spent training, especially next to batch methods that take forever. They looked at several online algorithms and checked how they do on accuracy, how well they adapt, and efficiency with ongoing data from sensors. It feels like incremental learning works great in smart sensor setups because data just keeps coming and the system has to respond fast. For gesture recognition, it lets the thing learn specific user habits over time, which makes it more personal and right on target. Plus, it scales up okay even in places that are changing a lot or short on resources. The experiments showed some of these incremental models beat the usual static ones when it comes to adapting in real time and staying steady. But there are issues too, like the model drifting off course, running out of memory, or figuring out how to mix old knowledge with the new without messing up. Overall, the research gives some good ideas on building adaptive systems for gestures and sensors. Continuous learning matters a ton in smart tech now, and I am not totally sure, but incremental stuff probably will be big for future human-machine interactions .

**Bimalkant Kachhatiya, Akash Patel, and Darshan Mehta. (2022)
'Implementation of TinyML Models on Arduino 33 BLE for Gesture &
Speech Recognition', arXiv preprint arXiv:2210.12345.**

This paper is about using models on an Arduino 33 BLE board. The main idea is that machine learning does not always need computers or cloud servers. TinyML models can be used on low-power embedded systems. This is important because sending data to the cloud can cause delays require internet connectivity and raise privacy concerns. The Arduino 33 BLE board has built-in sensors that the system uses. For gesture recognition it uses sensors like accelerometers to capture motion data when a user moves their hand. For speech recognition it uses a microphone to record voice commands. These raw signals are then processed using TinyML models. Since the hardware has limited memory and processing power the TinyML models must be carefully. Optimized to run efficiently. A major part of the paper focuses on optimization techniques. The authors mention methods like quantization, where model weights are reduced in size to make them fit into the memory. The workflow starts with training the machine learning models using datasets on more powerful systems like computers. Once trained these TinyML models are converted into a format for embedded devices and then deployed onto the Arduino using TensorFlow Lite for Microcontrollers. For gesture recognition the system detects motion patterns. Classifies them into predefined gestures. For speech recognition it listens for keywords or commands and triggers actions accordingly. Both processes run simultaneously. Provide quick responses, which shows that the system is capable of handling multiple tasks on a single microcontroller The experimental results show that the system achieves good accuracy while maintaining low power consumption.

Chen Zhao, Yu Du, and Xiaopeng Shao. (2022) ‘Real-time Interface Control with Motion Gesture Recognition via Non-contact Capacitive Sensing’, arXiv preprint arXiv:2212.02391.

This paper is about a way to recognize gestures using a special kind of sensing that does not need to touch anything. The main goal of this research is to make a system that lets people interact with machines without having to wear anything or be tracked by cameras. This makes things more convenient, cleaner and easier to use. The system is different from gesture recognition systems that use cameras or special sensors. This system uses a kind of sensing that can detect when a person's hand is near a surface. This lets the system figure out what gesture a person is making without them having to touch anything. The system is made up of an array of sensors that can detect when a person's hand is near. The system then uses computer programs to figure out what gesture a person is making. The people who made this system think that this kind of sensing is better because it uses power works fast and is not affected by things like light. This makes the system good for real-world applications like homes, public interfaces and healthcare. One of the things about this research is that the system can process gestures in real-time. The people who made this system used techniques to make sure that the system can respond quickly to gestures. This is important for applications that need to be interactive. This system uses a kind of sensing that can detect when a person's hand is near a surface. This lets the system figure out what gesture a person is making without them having to touch anything. The system is made up of an array of sensors that can detect when a person's hand is near.

The study also looked at kinds of gestures and how well the system could recognize them. The results of the study show that the system is very good at recognizing gestures even when people make them at distances or angles. The system can also work well with people without needing to be adjusted much. This makes it practical for a lot of people to use. The system is also durable because it does not have any moving parts. The paper also talks about how this kind of sensing can be used in a lot of devices like smartphones or smart panels. This could make it possible to make interfaces that use natural hand movements of buttons or touchscreens. The people who made this system think that combining this kind of sensing with kinds of sensing could make it even better. There are some challenges with this kind of sensing like it being affected by kinds of energy or not being able to detect complex gestures from far away.

Chih-Yang Lin, Meng-Chieh Wu, and Yi-Shiuan Liu. (2023) 'Simply Mime: A Control at Our Fingertips', Proceedings of the International Conference on Consumer Electronics (ICCE), IEEE.

This paper introduces a way to control devices called Simply Mime. It lets users interact with devices using hand movements. The main idea is to replace controls, buttons or touchscreens with a more intuitive and contactless interface. The authors want to make interactions feel more natural, faster and accessible in smart homes and consumer electronics. The system recognizes hand gestures. Converts them into control commands for devices. It uses sensors to capture user movements and applies machine learning to interpret those gestures. One of the strengths is that it reduces the need for physical interaction. This can be beneficial in situations where hygiene or conveniences important. It also helps simplify device control by reducing the complexity of remotes or interfaces into a single gesture-

based system. The Simply Mime system has stages. First it captures hand movements using sensors or cameras. Then it processes the signals to remove noise and extract features. The processed data is passed through a classification model that identifies which gesture was performed. Once recognized the system maps the gesture to a command, such as turning a device on or off. A significant contribution of this paper is its focus on real-time interaction with Simply Mime. The system responds quickly to user gestures ensuring an experience. The authors also consider usability aspects making sure that the gestures are simple and easy to remember. This is important because complex gestures can reduce user acceptance. The experimental evaluation demonstrates that Simply Mime can accurately recognize gestures with response time. The results show that gesture-based control can be an alternative to traditional interfaces. However the paper also points out challenges, such as variations in user behavior. The authors suggest that the Simply Mime framework can be extended to support gestures and additional devices. It also has use cases in areas like virtual reality and gaming. Overall this paper highlights the growing trend, toward natural user interfaces and smart interaction systems with Simply Mime. It shows how gesture recognition can simplify the way humans interact with technology making systems more intuitive. Future work may focus on improving accuracy and expanding gesture libraries for Simply Mime.

2.3 SUMMARY

Home automation and consumer electronics have changed a lot. We used to use remote control systems but now we have Internet of Things and software-based solutions. Before remote control systems used transmitters and receivers. Each device had its remote control and that was inconvenient for users. For example people had to use remote controls for their televisions, air conditioners and media systems. It was complex and inefficient. To solve this problem people made infrared remotes that could control many devices. Then smartphones became popular. People started using them to control devices. Smartphones can use Bluetooth, Wi-Fi. Infrared blasters to control devices. This meant that people did not need to use remote controls anymore. They could use their smartphones to control devices at the same time. Smartphone applications also made it easier for people to use their devices. They could customize the interface use voice control and access their devices from anywhere. In years microcontrollers like ESP32, ESP8266 and Raspberry Pi have changed the game. These microcontrollers are cheap. Can connect to the internet. They can also connect to cloud platforms so people can control their devices from anywhere in the world. This has led to the development of home ecosystems. In these ecosystems devices can talk to each other. Be controlled from one place. Cloud integration also allows for real-time monitoring, data logging and remote diagnostics. This makes these systems more intelligent and efficient. Nowadays smart systems are using sensors like Passive Infrared sensors and ultrasonic sensors. These sensors can detect presence and automatically turn devices on or off. This helps conserve energy. Ultrasonic sensors can detect proximity so people can use touchless interactions. This makes it safer and more convenient for people to use their devices. These sensor-based systems reduce the need for intervention and enable intelligent decision-making. Artificial Intelligence .

CHAPTER 3

METHODOLOGY AND WORKING PRINCIPLE

3.1 PROBLEM IDENTIFICATION

In our homes today we use things like TVs, air conditioners and sound systems. We control these things with controllers that send signals using infrared light.. We have so many devices now that we need a lot of controllers. This is a problem because it is hard to keep track of all of them and it can be confusing.

The old controllers also need us to touch them to work. We have to press buttons. This can be hard for old people or people who are not able to move around well. Sometimes we lose controllers or the batteries run out. The buttons can also be hard to understand and use. All of this makes it hard for us to use these devices easily.

There are some ways to control devices like using our phones or special smart home systems. These often need the internet to work and can be expensive or hard to set up. There are also systems that use cameras to see what we are doing with our hands.. These can be affected by things like bad lighting or noise in the room.

So we need an cheap way to control all of our devices without needing a lot of controllers. This way should not depend on things, like the internet or good lighting. It should also be easy to use. Not need us to touch a lot of buttons.

This project is trying to solve these problems by making a controller that we can wear. It will let us control our devices by moving our hands. This will make it easier and more convenient for us to use our devices

3.2 PROBLEM STATEMENT

We have a lot of devices at home like televisions and air conditioners. We use remote controls to turn them on and off. This can be a problem because we have to keep track of many remotes. It takes a lot of time. It can be very confusing. Some people, like people or people who are sick may have a hard time using these remotes because they have to touch them to make them work. This project is about making a kind of remote control that uses gestures. We will use sensors that we wear to detect the movements of our hands. Then we will use these movements to send commands to our devices.

The goal of this project is to make it easy and simple for people to control their devices. We want to make sure that everyone can use their devices no matter what. We think this will make life easier and more fun, for people. The gesture-based IR remote control system is what we are trying to make. We believe the gesture-based IR remote control system will be very helpful.

3.3 PRINCIPLE OF THE SYSTEM

The proposed system will incorporate a system of gesture recognition, infrared signals, and machine learning that will be used for the development of an intelligent universal remote control that will be able to control all the devices in the house using a single device. The objective of the proposed system is the development of a remotely controlled device that will replace all the existing remote controls and will be able to communicate with all the devices in a much simpler and efficient manner through the application of AI technologies. The developed system will be based on a microcontroller known as the ESP32, which will be the CPU of the entire system. All the hardware devices will be connected to the

microcontroller, and it will be able to communicate with all the hardware devices and control all the devices. The proposed system will be able to receive motion data from a motion sensor known as MPU6050, which is made up of two types of sensors, i.e., 3 axis accelerometers and 3 axis gyroscopes, and will be able to identify all the movements of the hands in the x, y, and z axes. The classification techniques of machine learning are incorporated in the gesture recognition system, i.e., KNN and Decision Tree. During the training phase, the gestures like swipe left and swipe right are collected from the MPU6050 sensor, which uses accelerometer + gyro technology, and are incorporated into the machine learning model for identifying the corresponding motion patterns for each of the gestures. After that, the machine learning model is implemented in ESP32 microcontroller, which enables it to identify the corresponding gesture with a high degree of accuracy. The next module of the system is the transmission and reception of IR signals for various functions of the remote control of the appliances. The IR Receiver module, when connected with the ESP32 module, is able to capture any IR signal sent from any of the Remote Control devices, i.e., Remotes of devices like televisions, air conditioners, projectors, and fans, and decode it, storing it in the memory of the microcontroller along with the function performed on the corresponding device, i.e., Power On/Off, Volume Up/Down, Channel Change, Temperature Up/Down, etc. Thus, the system is able to mimic the functionality of the original remote-control device without any modifications made to the devices themselves. Once the user performs the gesture while operating the system, the MPU6050 identifies the user's movement and sends the data related to the user's movement to the ESP32. Once the ESP32 receives the data related to the user's movement, the embedded machine learning algorithm will identify the type of gesture that matches the identified movement.

After that, the ESP32 selects one of the IR codes in its database that corresponds to the identified gesture and sends a signal to the appliance that corresponds to the identified gesture using an IR transmitter module. After receiving the IR signal sent by ESP32, the appliance will perform the action. Hence, the entire system is based on the concept of motion detection, intelligent gesture recognition, and IR communication. Thus, the user can control the appliances remotely using the smart remote controller without any physical contact with the appliance in real time. This provides the user with an efficient way of controlling their home or classroom automation system economically.

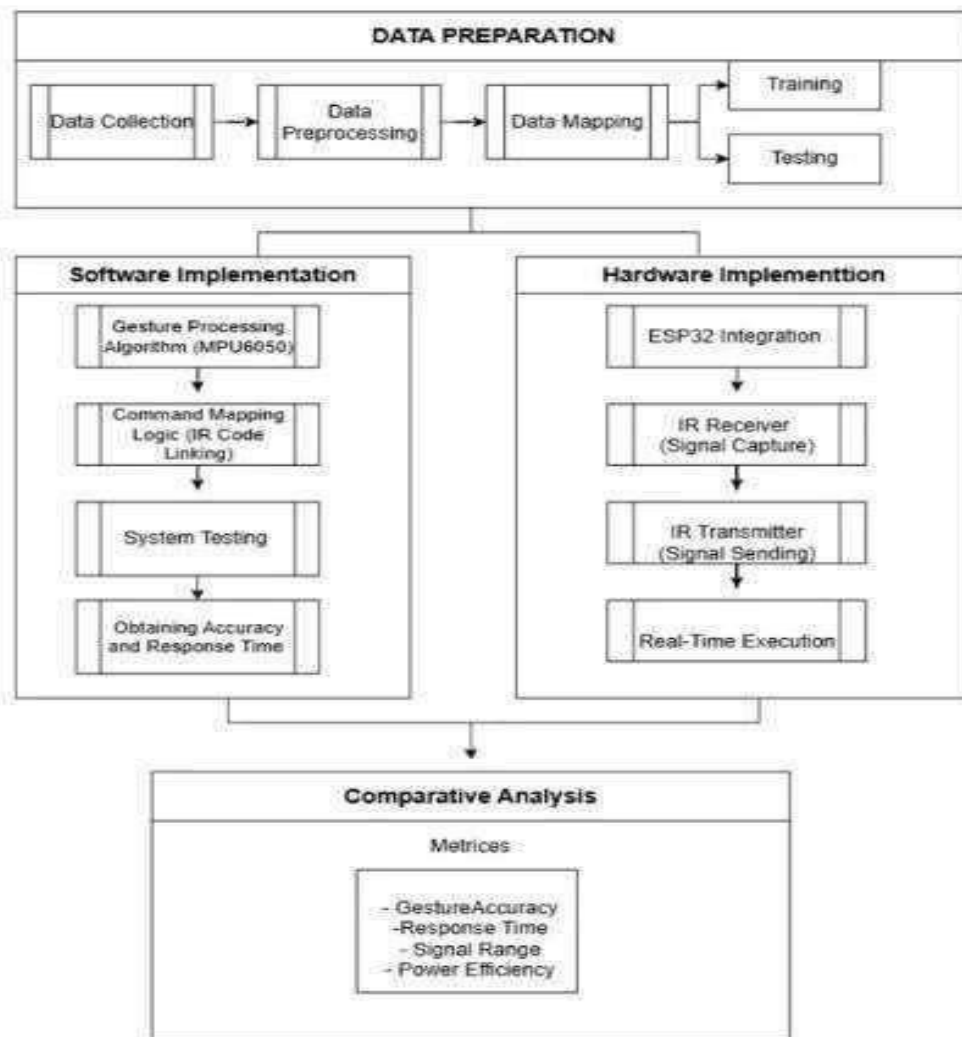


Figure3.1 Methodology of the proposed system

This proposed gesture recognition universal remote control utilizes machine learning techniques, including K-Nearest Neighbors (KNN) and Decision Tree, to achieve effective recognition of user gestures as well as control of household appliances accordingly. To start with, data is collected with the help of an MPU6050 sensor, collecting acceleration and gyroscope readings in relation to the three axes - X, Y, and Z. Afterwards, the collected data is processed by applying filters, extracting features such as direction, tilt angle, and other characteristics of the gesture. Feature extraction is considered to be an essential part of the gesture recognition process since it improves the quality of the input data. At this point, the processed data is further classified with the help of a machine learning algorithm known as K-Nearest Neighbors (KNN). KNN algorithm is a form of supervised learning which is used to classify data by comparing it with similar training data. This algorithm measures the distances between the new data and previous data points, then assigning it to the class shared by the majority of its nearest neighbors. Furthermore, in order to boost prediction speed and increase accuracy, the Decision Tree algorithm is also applied. The Decision Tree builds a tree-like structure, which consists of a series of decisions made according to the thresholds of features. Every node corresponds to a test result, and the leaf node will determine the identified gesture. This technique can make predictions quickly and easily understood. After the gesture has been categorized, the system will then translate it into a command that has been programmed into memory. The ESP32 microcontroller will retrieve the infrared (IR) code corresponding to the command and broadcast it using the IR transmitter. As such, the integration of KNN and the Decision Tree will ensure that the system recognizes gestures efficiently and accurately.

3.4 WORKFLOW OF THE PROPOSED SYSTEM

The developed AI-based universal smart remote system comprises different components that collaborate with one another in classifying the user's gestures. These components of the system include the following, which will form the workflow of the entire system: initialization of the system through the ESP32 microcontroller, gesture detection, classification of the detected gesture through machine learning techniques, mapping of the detected gesture with the corresponding infrared command, transmission of the command through infrared technology in controlling your electronic devices. The system can control various electronic devices in your entire house with only one device. The first step in the workflow of the entire system is the initialization of the system through the ESP32 microcontroller. Once the device is turned on, it initializes the hardware components connected with it in order to form the infrastructure of the user's gestures. These hardware components include the motion sensor device MPU6050, IR receiver module, IR transmitter module, and display. At this initial step of the system's workflow, the microcontroller loads the pre-programmed infrared codes corresponding with the user's actions, which have been trained using a machine learning technique in order for the microcontroller to recognize the user's gestures as intended. The first stage of the system is initialization, and once that is completed, we move on to the gesture sensing portion of our system. The user's hand is monitored by the MPU6050 sensor, which contains a three-axis accelerometer and a three-axis gyroscope. The accelerometer measures acceleration on the X, Y and Z axes (i.e.; three axes) while the gyroscope measures the rotational motion on the X, Y and Z axes. The data from both sensors is transmitted to the ESP32 (via the I2C protocol) in real-time to provide input for the

gesture recognition system. Data processing and gesture recognition is the next stage of the system. The ESP32 will filter out noise, extract useful motion features and use these filtered, usable, motion values as input into the embedded machine learning model which has been trained with K-Nearest Neighbors (KNN) and Decision Tree algorithms. The gesture recognition system will compare the incoming motion data against the already trained dataset to identify the gesture being made (i.e. swipe left, swipe right, move up or down). The gesture recognition process will be completed in real-time, allowing for rapid and accurate recognition of user gestures. The step after recognizing a gesture would be to identify the various commands that would be sent over to the respective appliances based on the relationship of the gesture with the command mapping. For example, a right swipe would power an appliance on, whereas a left swipe would power the same appliance off. The respective commands associated with the gesture commands would have an associated infrared code saved in memory based on the previously received infrared remote control signals sent to the respective appliances. If an infrared signal is sent over to the respective appliance, the appliance would react as if the original remote control was used. The next step after retrieving the respective infrared code associated with the gesture command would be for the ESP32 board to use its infrared transmitter module to transmit the code. The infrared transmitter would transmit a coded version of the original infrared code that was sent from the respective appliance's remote control. Since the infrared code saved in memory would be an exact replica of the code sent over from the original remote controls, the respective appliance would react as if the infrared code sent over from its respective infrared

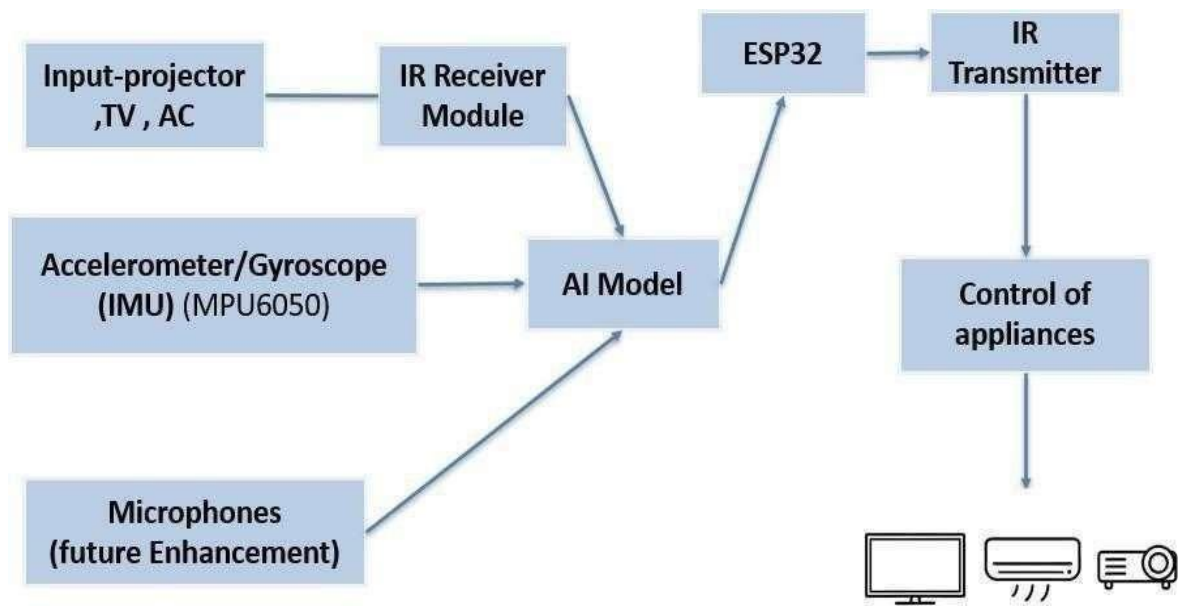


Figure3.2 Workflow of the proposedsystem

3.5 HARDWARE DESCRIPTION

For developing a gesture-controlled universal remote using embedded hardware components, the system will use an ESP32 microcontroller to control the entire process and communicate with devices using an infrared (IR) signal. For gesture detection, a motion sensor will be used, known as a MPU6050 motion sensor, which contains an accelerometer and a gyroscope to measure the orientation of the user's hand. The ESP32 will use the motion data from the MPU6050 motion sensor through an I2C interface. An IR receiver module will be used to receive IR signals from existing devices, and the IR code will be stored in memory using the ESP32. Once a gesture is detected, the ESP32 will use the IR code stored in memory to command a device such as a TV, air conditioner, projector, or fan using an IR transmitter module, an infrared light-emitting diode. The ESP32 will provide a user interface using an OLED display showing the detected gestures and the signals.

3.5.1 Microcontroller (ESP32)

The ESP32 microcontroller is the primary CPU of the proposed system. It receives the motion data from the MPU605 sensors. It identifies the gestures based on the trained machine learning model. It controls the emission of infrared (IR) signals from the system. It manages the communication between all the hardware modules. It enables the efficient control of the appliances in real time.



Figure3.3 Microcontroller (ESP32)

The other significant function of the ESP32 module is the reception and storage of the infrared codes sent by the original remote controller using an infrared receiver module. Each command sent to the appliance, such as ON or OFF, is stored in the memory of the ESP32 module as a specific hexadecimal number. Once the specific gesture is recognized, the ESP32 module sends the corresponding infrared code to the appliance using the infrared transmitter LED connected to it. Furthermore, the ESP32 module is responsible for the processing of sensor data in real time and facilitates fast communication between the input sensors and output devices. The ESP32 module is appropriate for the development of smart home and IoT-

352 MPU6050 Motion Sensor

In the proposed universal remote that enables gesture-based control of smart devices, the MPU6050 is an important motion tracking device that supplies motion data to identify the position of the hand.



Figure3.4 MPU6050 Motion Sensor

With the addition of linear acceleration and rotational motion, it is possible to use the MPU6050 to sense user gestures based on detected motion. In the proposed system for a gesture-controlled universal remote, the MPU6050 will be utilized to sense user hand movement and capture the acceleration and angular velocity values of each gesture made by the user.

The MPU6050 sends motion data to the ESP32 microcontroller. Therefore, since it utilizes the I2C protocol, it is able to send motion data at a quick rate. The motion data retrieved from the MPU6050 is then forwarded to the microcontroller for processing into input data for the (machine learning algorithm) of the proposed system for a gesture-controlled universal remote device.

3.5.3 Buck Converter

The proposed system of an AI-powered universal smart remote control would utilize a buck converter as a means of providing a stable power supply to the various individual components of the device. These individual hardware components of the device would include an ESP32 microprocessor module, an MPU6050 motion sensor module, an IR receiver module, an IR transmitter module, and an OLED display module. These components would need a specific and stable voltage supply in order to function optimally. If the device were using a higher voltage power source, such as a 9V or 12V battery or power adapter, the voltage would need to be stepped down to a safer and more suitable voltage level in order to power the device components.

The buck converter would step down the higher input voltage of the device to a lower output voltage of either 5 or 3.3 volts. These levels of voltage are safe and suitable enough to power the ESP32 module as well as all the other modules of the device. Linear voltage regulators have proven to be very inefficient in operation because of the dissipation of energy in the form of heat.



Figure 3.5 Buck Converter

3.5.4 IR Receiver Module

The IR receiver module is used in the proposed system in order to receive the infrared signals from the existing remote controls of the electrical appliances. This module is capable of detecting the infrared signals sent from the original remote controls of electrical devices such as TVs, air conditioners, projectors, and fans. These signals are received and transmitted to the ESP32 microcontroller.



Figure 3.6 IR Receiver Module

Each signal received has its own unique code associated with it that corresponds to a unique function of an electrical device, such as switching on/off a device, volume control, and temperature control, among many others. These IR signals are then decoded and stored in the memory of the ESP32 module for future use. Thus, through the reception and storage of these IR signals, the system is able to replicate the function of multiple devices using a single smart remote control device. The IR receiver is a device that is able to detect infrared light .

355 Power supply / Battery

The power supply has an important role to play in the proper functioning of the universal smart remote control system. It has to provide the required electrical power to operate all the hardware components of the universal smart remote control system, which includes the ESP32 microcontroller, the MPU6050 motion sensor, the IR transmitter, the IR receiver, and the OLED display. In this project, the required power supply can be given by a USB power source or a rechargeable battery. If the power source is given by a USB source, it is required to provide 5 volts of electrical power using a laptop or a power source. The voltage is internally regulated to 3.3 volts by the ESP32 module, as it is the required voltage to operate



Figure 3.7 Battery

the entire system. The required power can also be given by a rechargeable battery. This is because it enables the smart remote to function even in the absence of a physical connection, thus allowing it to be carried around and used in various locations, including homes and classrooms. The power supply enables the system to function and operate effectively.

35.6 Infrared Transmitter

The infrared transmitter module is one of the key components of the universal smart remote control system. This helps in controlling numerous devices, e.g., TVs and fans, etc. For our project, we are utilizing the transmitter module in conjunction with the ESP32 microcontroller. The microcontroller, in this case, is like the person in charge of this project. The microcontroller, in this case, receives information from the MPU6050 sensor. After that, it looks at the information. Then, it sends a command. After that, the microcontroller sends a code to the transmitter module. The infrared transmitter module sends out a signal that is similar to the one that a normal remote control sends to a device. This helps in controlling numerous devices without interfering with anything else. The infrared



transmitter module helps in controlling numerous devices with just one device. The infrared transmitter module is a key component of this system. It helps the microcontroller in controlling the devices.

Figure 3.8 IR Transmitter

35.7 OLED Display

OLED display is used in the proposed smart remote system. It will assist in understanding the actions being performed by the system. It will also assist in understanding the detected gesture and selected command to operate the appliances. In the proposed smart remote system, the OLED display is used along with the ESP32 microcontroller. It will send information to be displayed on the OLED screen. Once the gesture is detected by the MPU6050 sensor, information is processed by the ESP32 microcontroller. Then, the command is recognized by the microcontroller. The user is informed about the recognized gesture and selected command by using the OLED display. It will show information to the user that the system is able to understand the detected gesture. OLED display is used in various embedded systems due to their bright display, low power consumption, and clear visibility. Due to the small size of the OLED display, it is suitable for using in portable devices such as smart remote. With the use of the OLED display, the overall usability and interaction of the proposed smart remote have been enhanced.



Figure 3.9 OLED Display

3.6 SOFTWARE DESCRIPTION

The software system is the major controlling part that controls the entire process of the smart universal remote. The software has been developed using the C/C++ programming language with the Arduino Integrated Development Environment (IDE) and the ESP32 microcontroller. The system receives the constant information from the MPU6050 sensor. The sensor identifies the different hand gestures made by the user. After that, the system identifies the different hand gestures made by the user using the K-Nearest Neighbor (KNN) and Decision Tree algorithms. After identifying the hand gesture defined by the user, the system maps the hand gesture with the command for operating the home appliances. After that, the system allows the infrared transmitter to transmit the required infrared signal for operating the home appliances such as TVs, air conditioners, projectors, and fans. The system processes the information in real-time and minimizes the memory usage for identifying the hand gestures made by the user.

3.6.1 Embedded Firmware (ESP32 Code – Arduino IDE)

The microcontroller of the ESP32 is used and programmed with a C/C++ compiler in the Arduino Integrated Development Environment to develop the core functionality of the smart universal remote control system. The firmware is able to continuously read motion data from the MPU6050 motion sensor and process it to identify the user gestures in real time. Depending on the identified gestures, the ESP32 sends the stored infrared signal to control different appliances like TVs, air conditioners, projectors, and fans. The firmware is also able to manage the interactions between the gesture recognition module, the IR transmission module, and the display unit to ensure the smooth and reliable functionality .

3.6.2 Gesture Recognition Software

The implementation of the gesture recognition module is performed by using machine learning algorithms. These machine learning algorithms help in the precise recognition of hand gestures. For this purpose, the data is collected by the motion sensor MPU6050. The data is then processed and analyzed to recognize hand gestures such as swipe left, swipe right, move up, and move down. These machine learning algorithms are chosen because they are simple and accurate for embedded systems. After the recognition of the gesture by this module, it is mapped with a particular appliance.

3.6.3 Data Processing and Model Training

The data collected by the sensor is processed, and the training of the machine learning model is carried out. The training of the machine learning model is done with the help of Google Colab. Google Colab is a cloud-based platform where data analysis is carried out. The data is pre-processed to remove noise from the data. This is an important step in the process. The optimization of the machine learning model is carried out so that it runs on the ESP32 microcontroller.

3.6.4 Communication and Control Mechanism

Communication among the various modules of the system is carried out internally by the ESP32 microcontroller. The MPU6050 sensor uses the I2C protocol for efficient data transfer with the ESP32 microcontroller. After the processing of the gestures, the corresponding infrared code is retrieved and sent wirelessly by the IR transmitter module. This allows for the wireless control of various home appliances without original device.

3.6.5 User Interface and Display Feedback

The system also includes an OLED display module that is used as a simple user interface. The display provides real-time feedback by indicating the detected gesture and the selected command. This is very important in the system because it helps the user confirm whether the system has correctly interpreted the performed gesture. The display makes the system more interactive and user-friendly.

3.6.6 System Monitoring and Execution

Home automation and consumer electronics have changed a lot over time. We used to have remote control systems that relied on special hardware. These systems used signals to control devices like televisions and air conditioners. Each device needed its remote control, which was really inconvenient for users. People had to deal with a bunch of remotes. It was hard to manage them all. Then something new came along. We started using remote controls that could work with lots of different devices. These remotes could learn the signals from remotes, so you only needed one remote to control everything. This was an improvement, but it was still not perfect. With the rise of smartphones things started to change more. We began to use our phones as controls. This was possible because phones have things like Bluetooth and Wi-Fi built in. We could use apps to control our devices. It was a lot more convenient than using a bunch of separate remotes. These apps also had some cool features, like voice control and customizable interfaces.

3.7 INNOVATIVENESS OF THE SOLUTION

The proposed design for the AI-powered universal smart remote proposes a unique solution by incorporating gesture recognition technology and IR communication in a single system. The developed system is different from other smart remote control systems in that it does not require users to manually use a remote control by pushing a button. Instead, it allows users to control various appliances within a home or a classroom by simply performing hand gestures. The proposed system incorporates the ESP32 microcontroller module that allows real-time processing and system control within a low-cost platform. The proposed system uses a motion sensor MPU6050 to sense user gestures and incorporates machine learning algorithms such as K-Nearest Neighbors and Decision Tree to classify user gestures and provide accurate results. The proposed system is efficient and user-friendly compared to other smart remote control systems that require users to use multiple remote control devices and may not be compatible with various appliances within a home or a classroom due to a lack of IR learning capabilities. The proposed system incorporates an OLED display that provides real-time feedback and increases user confidence in using the proposed system.

3.8 IMPACT OF THE PRODUCT

The system eliminates the need for multiple remotes in favor of a single smart system, making it easy for users to control their appliances through gestures. The system is developed using components such as ESP32 and is cost-effective. This makes it possible for the system to be expanded to control multiple devices, making it useful in homes and learning institutions. The system incorporates gesture recognition technology, IR communication technology, and machine learning technology.

3.9 UNIQUENESSAND FEATURE

The system lets you control devices without touching them using hand gestures that the MPU6050 sensor can detect. This sensor keeps track of your motion and orientation all the time so it can figure out what gesture you are making without you having to touch anything. This makes things easier and more convenient in your life.

You can use this system to control a lot of devices like your TV, air conditioner, projector and fan all with one system. This means you do not need a lot of remote controls and you can control everything from one place. The system uses computer programs, like KNN and Decision Tree to understand what gesture you are making. It looks at the data from the sensor in time so it can be more accurate and reliable. It can also. Store the signals from other remote controls so it can work with a lot of different devices from different brands. The ESP32 microcontroller makes the system work fast and well and it has built-in Wi-Fi and Bluetooth so you can connect it to devices easily. When you make a gesture the system gives you feedback away so you can understand what it is doing and use it more easily. The system is small. Can fit in your home or classroom and you can add more devices and features to it later so it can keep up with new technology. The system is very adaptable and reliable. It can work with a lot of different devices making it a great tool and a innovative tool, for the future.

3.10 SUMMARY

The history of universal control systems has come a long way since the simple, single-device transmitter to the highly advanced, AI-based hub that now controls the entire smart home. The research was initially directed at resolving the major problem of "remote clutter" by developing a single handheld device that could decode and store thousands of proprietary infrared protocols. The initial universal remote control, which uses an 8-bit processor such as the Samsung S3F80PB or the Texas Instrument MSP430, was developed with the primary goal of compiling the codes of television sets, air conditioning units, and DVD players into a single database, thereby eliminating the wastage of resources. As the usage of mobile devices has increased, research was directed at utilizing the smartphone as the primary interface for the universal control systems. By using Android-based applications and protocols such as Bluetooth and Wi-Fi, the limitation of the number of buttons was overcome by the development of the interactive touchscreen interface, thereby allowing the user to have "anytime, anywhere" control through the Internet of Things (IoT) by utilizing the highly advanced processor, Raspberry Pi 3, for data fusion and access through the cloud. The most recent developments have also enabled the inclusion of artificial intelligence and gesture recognition in the control system. This involves incorporating the ESP32 microchip and motion sensors like MPU6050. Moreover, this involves the utilization of machine learning algorithms to ensure that the control system recognizes hand gestures via the utilization of the K-Nearest Neighbor (KNN) algorithm. This ensures that the user of the control system does not need to physically interact with it. This is due to the fact that a swipe of the hand would be enough to control lighting and temperature in a room, thus making it a very intuitive control system.

CHAPTER 4

DESIGN AND MODELLING

4.1 DESIGN

The proposed system is to be designed as an "AI-enabled Universal Smart Remote" that includes all sorts of control systems in one single device. The main component of the system includes an ESP32 microcontroller, which will be the main processor and communication center. The system also includes an MPU6050 accelerometer and gyroscope sensor to detect hand gestures. These gestures are then processed and input to a light-weight AI, which then recognizes and processes these gestures as corresponding commands, such as power ON and power OFF.

The IR receiver module is used in the first stage to store the hexadecimal values of the existing remote controllers. These hexadecimal values are transmitted with the help of an IR transmitter module (IR LED) to control various home appliances such as TV, air conditioners, projectors, fans, etc.

The system architecture is designed in such a manner that it can be extended to incorporate voice control with the help of microphone sensors and IoT remote access in the future. The system design is aimed at providing a cost-effective, user- friendly, and intelligent solution that can enhance the level of convenience by replacing the need for multiple remotes with a single smart system for all operations.

The system allows for quick response and accurate performance with minimal delay. The system is made to be flexible and expandable for future enhancements. Additional features such as voice control and IoT connectivity can be included in the system. This has made the system more user-friendly and appropriate for a smart home and smart classroom environment

Hardware elements, software applications, as well as AI are used to facilitate the operation of the system and its efficiency. The system's main architecture consists of three main components: the user input component, the processing component, and the appliance control component. The user may control the entire system with the help of mobile applications or voice commands. The core of the system is made up of the processing module, which is comprised of the microcontroller as well as AI technologies, which interpret the user's commands

This type of system is smart and can learn from user's behavior and automate certain actions in order to increase the efficiency of the system as a whole. After the processing is done, the command is sent to the respective device through the communication protocol (IR, Bluetooth, and Wi-Fi). The traditional devices such as TV sets can be controlled with the help of IR communications, while other smart devices should be connected via Bluetooth or Wi-Fi

As discussed above, the design of the system comprises embedded systems, wireless communication and artificial intelligence. The heart of the system is the controller or microprocessor, functioning as a control unit. This controller will get data from an appliance and perform tasks related to controlling the appliance based on the instructions received from the user. The design will include software which would be an interface for the user to control his/her devices. This portion of the system

needs to be intuitive and convenient for the user to operate, allowing the selection of the appliance to give it commands or check its current status. The one of the greatest benefits of this design is the usage of the artificial intelligence function in this system. Being equipped with artificial intelligence, the system recognizes actions of users and performs similar tasks automatically in the future without the need for manual input from the user. For instance, when the user regularly turns on the fan during a particular hour of the day, or sets a certain temperature for an air conditioner, the system will recognize this routine and repeat the task in the future automatically. Communication between the control unit and the home appliances will be carried out through a number of ways that include Infrared, Bluetooth, and Wi-Fi communications. While Infrared communication technology will be used to control the non-smart home appliances, which may include televisions and set-top boxes, Bluetooth and Wi-Fi technologies will be used to control the smart home appliances. This will ensure that one solution can be used to control both old and new home appliances. The other key feature is scalability, which is another important consideration in the system design. The way the system has been designed ensures that further home appliances can be added to be controlled without affecting the existing system. Security considerations have been made in the system through incorporating some level of user authentication within the mobile application.

Reliability and real-time processing are other factors apart from functionality that have been taken into consideration while designing this project. This system can process command immediately without any delay. In addition, there can also be error-detection mechanisms incorporated which will help detect errors that might arise during the process of communication or execution of commands.

4.2 EXPERIMENTAL STUDY

An experimental case study involving the designing of an artificial intelligence-based universal remote for seamless control of home appliances has been carried out under the title of the project. The reason for conducting the experiment was to determine the feasibility of the project design and ensure that it can perform efficiently. Therefore, the aim of the experiment is to prove that the designed model will be able to work as an effective substitute for the traditional remotes in controlling different devices. The process of carrying out the experiment involved constructing a model of the product. The designing of the device involved the use of microcontrollers as well as several communication platforms including infrared waves, Wi-Fi, and Bluetooth. Additionally, a mobile application was created as the means through which the user would interact with the product. In terms of the experimentation environment, a room was created. Also, devices including a TV set, a fan, the lighting system, and a smart plug were chosen for the experiment. The following tests were carried out in order to check the efficiency of the system from several sides. To begin with, basic tests were performed concerning the overall functionality of the system. Some instructions were delivered to the system through the mobile app via turning certain appliances on/off, regulating the fan's speed, and choosing TV channels. It has been found that all those instructions were successfully delivered, analyzed in the central controller, and the necessary actions have been made in terms of sending appropriate signals to particular devices. In addition, the time needed to carry out these instructions was very little as most of the actions were fulfilled right away.

Apart from that, the efficiency of communication modules was checked separately in order to see how good they were. Testing the communication via infrared signals was done with the help of the TV. In this case, it was established that all the signals could be identified by the system and appropriately analyzed in order to carry out different functions related to switching, controlling the volume level. Thirdly, it was related to the examination of AI-based characteristics of the tool. The program was instructed to monitor the behavior of the users for a few days. Then, the application examined the data, including popular devices, preferences, and the time when they were utilized. Based on this data, the algorithm could recommend some actions to be undertaken automatically for the users, such as turning off or on certain devices according to their usual activity at particular times. If the user always turned on the fan at the same time every evening, then it either recommended or performed this action without user's involvement. Usability and user experience were the critical criteria for the assessment of the case study. The set of people with diverse knowledge about technology was invited to operate the system during the experiment. These individuals were instructed on the tasks that needed to be completed using the software and the voice command. The responses were collected through observation and surveys. The vast majority of participants found the utilization of the app extremely convenient. It was pleasant to control the home appliances with the help of a single application. The various kinds of interruptions were created to conduct reliability tests, and the reactions of the system were observed. Network interruptions were created when there was transmission of data through Wi-Fi connections. Whenever such an

notify the user of its failure. In another instance, wrong directions were given to the system. Saving on energy was one more area that was examined in this study. This could be achieved by evaluating the manner in which appliances were used and automating certain actions. Appliances that were turned on unnecessarily could be switched off by the system after a period. Energy consumption would be reduced as a result, while extending the lifespan of the appliance. Artificial intelligence ensured that appliances used energy efficiently. Disadvantages associated with the experimentally conducted study were identified as well. During the first stages of work, the system cannot deliver exact results or automate operations. Also, there could be some issues related to the compatibility of devices since some of them do not possess a communication protocol. If Wi-Fi connection is used to provide control over the system, then a robust network should be established. To achieve further developments, machine learning algorithms need to be enhanced and extended. In addition, the database of devices needs to be enlarged. The use of cloud technology will enable the remote control and operation of the system from anywhere in the world. Finally, security measures need to be implemented. In conclusion, according to the conducted experiment, it can be stated that the developed universal intelligent remote control system with the application of artificial intelligence technology works effectively in practice. Moreover, the introduction of a number of innovative technologies to the same device turned out to be a successful decision in regard to designing an efficient intelligent remote control for different home appliances.

43 CODE

```
#include <Wire.h>
#include <U8g2lib.h>
#include <IRremoteESP8266.h>
#include <IRsend.h>
#include <ir_Samsung.h>

// ===== PINS =====

#define IR_LED_PIN 18
#define BTN_UP 33
#define BTN_DOWN 25
#define BTN_SELECT 32

// ===== OBJECTS =====

IRsend irsend(IR_LED_PIN);
IRSamsungAc ac(IR_LED_PIN);

U8G2_SSD1306_128X64_NONAME_F_HW_I2C
u8g2(U8G2_R0, U8X8_PIN_NONE);

// ===== MENU =====

enum Page { MAIN_MENU, SUB_MENU };
Page currentPage = MAIN_MENU;
const char* devices[] = {"SONY TV", "DTH", "SAMSUNG AC"};
int deviceIndex = 0;
int signalIndex = 0;
```

```

// ===== SIGNAL LISTS =====

// ---- SONY TV ----

const char* sonySignals[] = {
    "POWER", "VOL +", "VOL -", "CH +", "CH -",
    "MENU", "UP", "DOWN", "LEFT", "RIGHT"
};

// ---- DTH ----

const char* dthSignals[] = {
    "POWER", "VOLUME", "CHANNEL"
};

// ---- AC ----

const char* acSignals[] = {
    "ON", "OFF", "TEMP +", "TEMP -"
};

// ===== AC STATE =====

bool acPower = false;
uint8_t acTemp = 24;

// ===== TIMING =====

unsigned long lastActionTime = 0;
#define RETURN_TIME 20000
#define DEBOUNCE 300

```

```
//===== DRAW =====

void drawMainMenu() {
    u8g2.clearBuffer();
    u8g2.setFont(u8g2_font_ncenB08_tr);
    u8g2.drawStr(0, 12, "Select Device");

    for (int i = 0; i < 3; i++) {
        if (i == deviceIndex) u8g2.drawStr(0, 30 + i * 10, ">");
        u8g2.drawStr(12, 30 + i * 10, devices[i]);
    }
    u8g2.sendBuffer();
}

void drawSubMenu() {
    u8g2.clearBuffer();
    u8g2.setFont(u8g2_font_ncenB08_tr);
    u8g2.drawStr(0, 12, devices[deviceIndex]);

    const char** list;
    int count;

    if (deviceIndex == 0) { list = sonySignals; count = 10; }
    else if (deviceIndex == 1) { list = dthSignals; count = 3; }
    else { list = acSignals; count = 4; }

    for (int i = 0; i < count; i++) {
```

```

    if (i == signalIndex) u8g2.drawStr(0, 28 + i * 8, ">");
    u8g2.drawStr(12, 28 + i * 8, list[i]);
}
u8g2.sendBuffer();
}

// ===== SEND FUNCTIONS =====

void sendSony(uint16_t code) {
    for (int i = 0; i < 3; i++) {
        irsend.sendSony(code, 12);
        delay(40);
    }
}

void sendDTH(uint64_t code, uint8_t bits) {
    irsend.sendRC6(code, bits);
}

// ===== ACTION =====

void sendSignal() {

    // ===== SONY TV =====
    if (deviceIndex == 0) {
        Serial.print("SONY: ");
        Serial.println(sonySignals[signalIndex]);
    }
}

```

```

switch (signalIndex) {
  case 0: sendSony(0xA90); break; // POWER
  case 1: sendSony(0xA50); break; // VOL +
  case 2: sendSony(0x2F0); break; // VOL -
  case 3: sendSony(0x490); break; // CH +
  case 4: sendSony(0xC90); break; // CH -
  case 5: sendSony(0xA50); break; // MENU
  case 6: sendSony(0xAF0); break; // UP
  case 7: sendSony(0x2D0); break; // DOWN
  case 8: sendSony(0xCD0); break; // LEFT
  case 9: sendSony(0x2F0); break; // RIGHT
}
}

// ===== DTH =====

else if (deviceIndex == 1) {
  Serial.print("DTH: ");
  Serial.println(dthSignals[signalIndex]);

  if (signalIndex == 0) sendDTH(0xC8041A70C, 36);
  if (signalIndex == 1) sendDTH(0xC80412711, 36);
  if (signalIndex == 2) sendDTH(0x1900834E, 29);
}

```

```

else {
    Serial.print("AC: ");
    Serial.println(acSignals[signalIndex]);

    if (signalIndex == 0) acPower = true;
    if (signalIndex == 1) acPower = false;
    if (signalIndex == 2 && acTemp < 30) acTemp++;
    if (signalIndex == 3 && acTemp > 18) acTemp--;

    if (acPower) {
        ac.on();
        ac.setMode(kSamsungAcCool);
        ac.setTemp(acTemp);
        ac.setFan(kSamsungAcFanAuto);
    } else {
        ac.off();
    }
    ac.send();
}

// ===== SETUP =====

void setup() {
    Serial.begin(115200);

```

```

pinMode(BTN_DOWN, INPUT_PULLUP);
pinMode(BTN_SELECT, INPUT_PULLUP);

u8g2.begin();
irsend.begin();
ac.begin();

drawMainMenu();
Serial.println("UNIVERSAL REMOTE READY");
}

// ===== LOOP =====

void loop() {

  if (currentPage == SUB_MENU &&
      millis() - lastActionTime > RETURN_TIME) {
    currentPage = MAIN_MENU;
    drawMainMenu();
  }

  if (digitalRead(BTN_UP) == LOW) {
    delay(DEBOUNCE);
    lastActionTime = millis();

    if (currentPage == MAIN_MENU && deviceIndex > 0) deviceIndex--;
  }
}

```

```

    currentPage == MAIN_MENU ? drawMainMenu() : drawSubMenu();
}

if(digitalRead(BTN_DOWN)==LOW) {
    delay(DEBOUNCE);
    lastActionTime = millis();

    int max = (deviceIndex == 0) ? 10 : (deviceIndex == 1 ? 3 : 4);

    if(currentPage == MAIN_MENU && deviceIndex < 2) deviceIndex++;
    else if(currentPage == SUB_MENU && signalIndex < max - 1)
        signalIndex++;
    currentPage == MAIN_MENU ? drawMainMenu() : drawSubMenu();

    if(digitalRead(BTN_SELECT)==LOW) {
        delay(DEBOUNCE);
        lastActionTime = millis();
        drawSubMenu();
    } else {
        sendSignal();

        drawSubMenu();
    }
}
}
}

```

4.4 SUMMARY:

The proposed Universal Smart Remote device with AI features is intended to simplify modern living by connecting various smart controllers into a single smart controller. The Universal Smart Remote device utilizes a microcontroller known as ESP32. The microcontroller plays a crucial role as the brain of the Universal Smart Remote device. The microcontroller is utilized to process and communicate information. To improve the Universal Smart Remote device and advance beyond the fundamental idea of pressing buttons, the MPU6050 accelerometer and gyroscope sensor have been incorporated into the Universal Smart Remote device. The MPU6050 sensor enables the Universal Smart Remote device to recognize hand gestures, and the light AI performs various tasks by interpreting the hand gestures. The Universal Smart Remote device has two stages of operation. The first stage involves utilizing the IR receiver to store the hexadecimal code from various remotes. The second stage involves utilizing the IR LED transmitter to transmit the hexadecimal code to various devices such as TVs, air conditioners, and projectors.

CHAPTER 5

EXPERIMENTAL RESULTS AND DISCUSSIONS

5.1 RESULT FOR THE PROPOSED SYSTEM

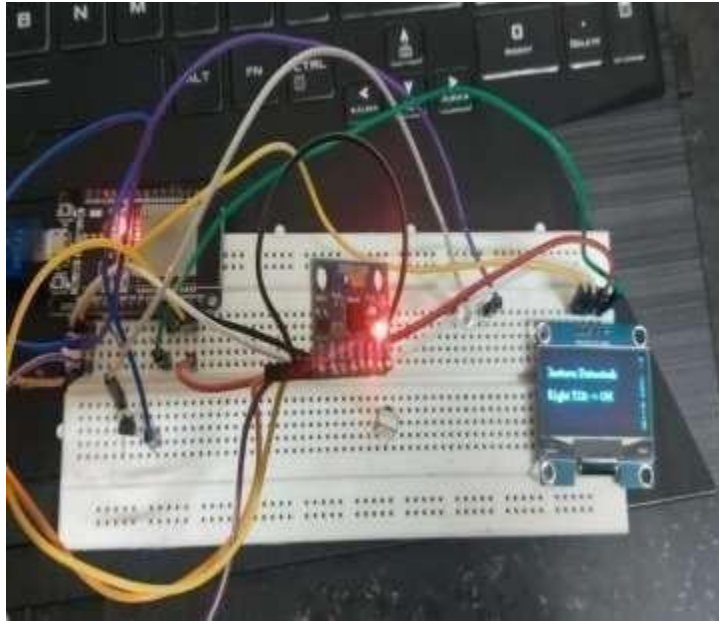


Figure 5.1 Hardware Setup and Implementation

figure 5.1 shows the experimental setup of a gesture-based control system built using an ESP32 microcontroller, MPU6050 accelerometer and gyroscope sensor, and a 0.96-inch OLED display. The components are connected on a breadboard using jumper wires, with communication established through the I2C protocol. The MPU6050 detects motion and tilt gestures, which are processed by the ESP32 in real time. The OLED display provides visual feedback, showing the message “Gesture Detected: Right Tilt -> ON,” confirming successful recognition of the gesture. The glowing LED indicates active operation, demonstrating that the system accurately converts physical gestures into control commands.

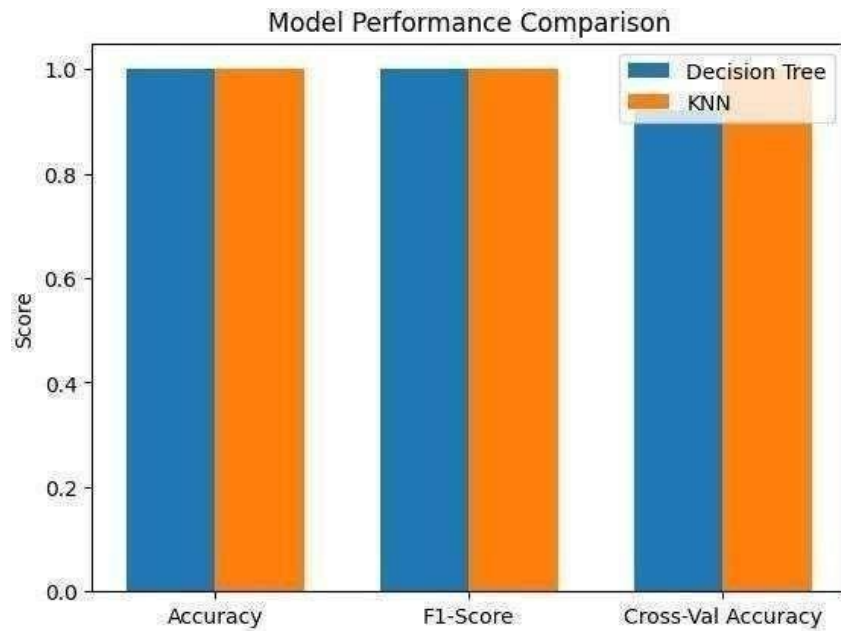


Figure 5.2 Comparison of the algorithm

figure 5.2 shows the performance comparison between Decision Tree and KNN models using accuracy, F1-score, and cross-validation accuracy. Both models achieve a perfect score of 1.0 in all metrics. This indicates that both algorithms perform equally well on the given dataset, with high classification accuracy and consistency.

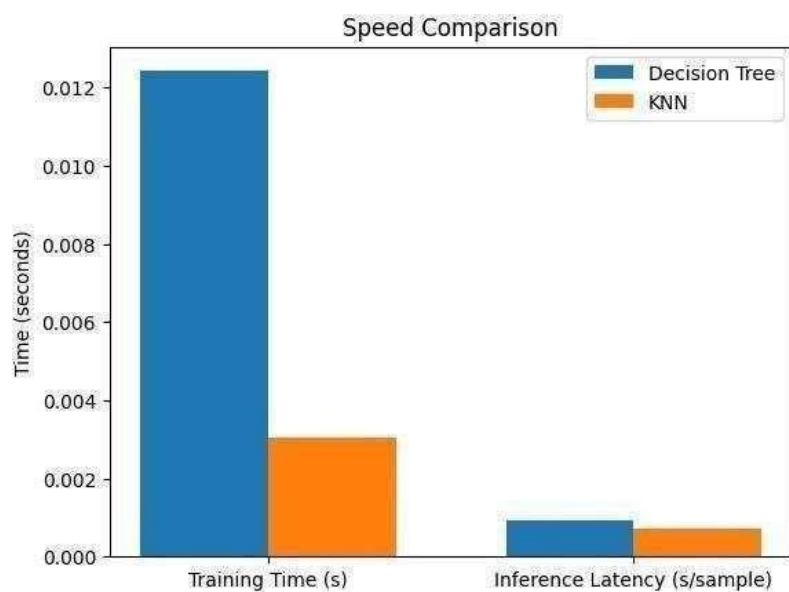


Figure 5.3 Speed Comparison of the algorithm

figure 5.3 shows the speed comparison between Decision Tree and KNN models in terms of training time and inference latency. KNN has lower training time compared to Decision Tree, while both models exhibit very low inference latency. However, KNN performs slightly faster overall, indicating better efficiency in this implementation.

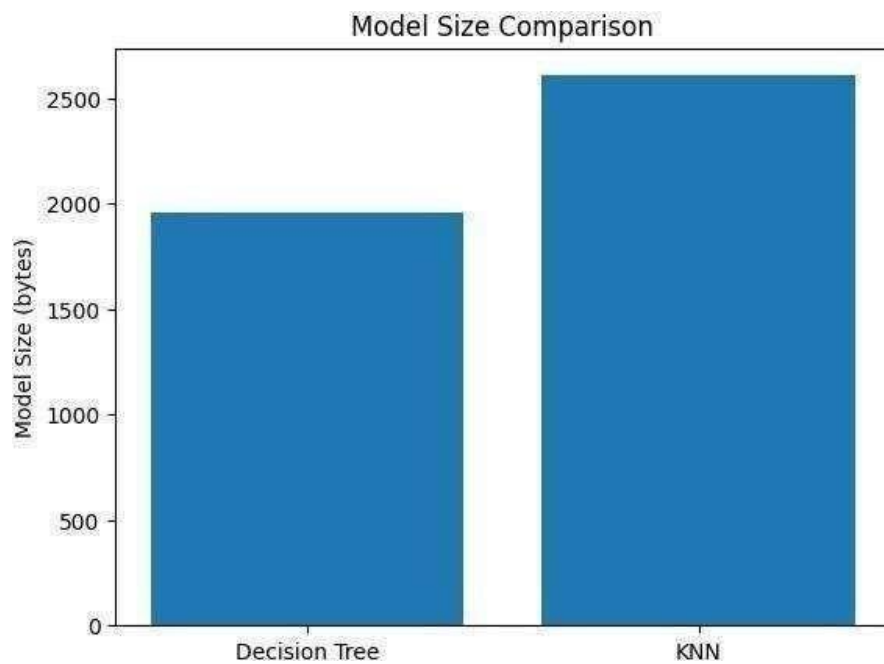


Figure 5.4 Model size Comparison of the algorithm

Figure 5.4 shows the model size comparison between Decision Tree and KNN algorithms. The Decision Tree model has a smaller size of approximately 1960 bytes, while the KNN model is larger at around 2620 bytes. This indicates that the Decision Tree is more memory-efficient, making it more suitable for resource-constrained embedded systems.

ML Model Comparison Report

Metric	Decision Tree	KNN
Accuracy	1.000	1.000
Precision	1.000	1.000
Recall	1.000	1.000
F1-Score	1.000	1.000
Cross-Val Accuracy	0.920	1.000
Training Time (s)	0.012	0.003
Inference Latency (s/sample)	0.001	0.001
Model Size (bytes)	1958	2608

Table 5.1 comparison of the algorithm

The table 5.1 presents a comprehensive comparison between Decision Tree and KNN models across multiple performance metrics. Both models achieve perfect scores of 1.0 in accuracy, precision, recall, and F1-score, indicating excellent classification performance. KNN slightly outperforms in cross-validation accuracy with a perfect score, while Decision Tree shows a marginally lower value. In terms of speed, KNN requires significantly less training time and has lower inference latency. However, the Decision Tree model is more memory-efficient, with a smaller model size compared to KNN. Overall, KNN offers better speed and consistency, while Decision Tree is more suitable for memory-constrained systems. Both KNN and the Decision Tree algorithms are highly effective in evaluating the data from the current case study. The former algorithm would be preferable in situations where speed and efficient use of memory resources are critical factors. In contrast, the latter approach would be more appropriate if memory usage and interpretability are key considerations.

5.2 SUMMARY

From the above results, one can see the difference in performance of the Decision Tree and KNN model utilized by us. The cause behind it is that both the models achieved perfect scores in accuracy, precision, recall, and f1 score, which were equal to 1.0. From this, one can conclude that both the classifiers classified all the cases present in our dataset accurately, without making any errors. Thus, it can be said that both the algorithms performed exceptionally well because of the properly prepared data set.

Other than performance measures, cross-validation was also carried out in order to examine the generalization ability of the models of machine learning. From the analysis, it is evident that KNN performed a little bit better than the decision tree when the cross validation accuracy criterion was used. Therefore, it can be inferred that KNN is consistent and is less vulnerable to overfitting in the present scenario. The reason behind this is that KNN is a distance-based classifier and therefore, will do well if the same class of data points are near each other. It appears that the current data set fits into this category. From the aspect of computation, it can be observed that there is a clear distinction between the two models. The training time for KNN was extremely low due to the reason that KNN is a lazy learner and therefore, does not need any modeling at all during training. As a result, the training data is preserved and computing takes place during testing. Although it is a lazy learner, the prediction time was low due to the fact that the number of instances was manageable. Moreover, the Decision Tree approach is also superior when it comes to interpretability, since the decision-making process can be interpreted =

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The development of the AI-Enabled Universal Smart Remote is a groundbreaking invention in automation for both homes and classrooms, as it has successfully addressed the challenge of handling different controllers for different brands. Unlike other conventional remotes that only utilize physical buttons for their operation, this system has successfully integrated an MPU6050 motion sensor for detecting the orientation and movement of the user's hands in real time. However, this is not enough for the system to function optimally, as there is also a need to translate this physical movement into digital language, and this is done by using machine learning algorithms such as K-Nearest Neighbors (KNN) and Decision Trees. Another very critical factor of this project is that it is universally compatible, given the presence of a highly advanced infrared learning module. This module has used an IR receiver to detect and store the unique hexadecimal signal patterns of any existing remote control. This allows the device to replicate any existing remote control, ranging from television sets, air conditioners, and even projectors. This data is sent back through an IR LED once a particular gesture is detected. Furthermore, a user-friendly experience is guaranteed through the use of an OLED display, which provides immediate feedback regarding the detected gesture and the command, thus greatly reducing the possibility of user error. Overall, this is a highly scalable, cost-effective, and miniaturized solution .

6.2 FUTURE ENHANCEMENT

The project can be made better by adding voice control. This means users can use voice commands to operate their appliances. We can do this by adding speech recognition or cloud-based voice assistants. This way users can control devices without using their hands, which's easier and more accessible. The project can also be made better by connecting it to the internet using IoT. This way users can operate their appliances using their phones from anywhere in the world. This allows users to monitor, schedule and automate devices from afar making the system more flexible and convenient for users.

The project can also be made better with an app. This makes the project more user-friendly. A special app can give users an interface controls that they can customize and updates on the status of their devices in real time. This makes the overall experience better for users. The project can also be improved in the future to learn what users like and do things on its own. By using machine learning the system can look at how users use it and do things automatically based on what users like. This makes the system more efficient. Reduces the need for users to do things manually. The gesture recognition can also be made better using cameras. This gives accurate results. Systems that use cameras can detect complex gestures and are more precise than systems that use sensors.

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SDG Mapping



The project titled **AI-Powered Universal Smart Remote for Gesture-Based Control of Home Appliances** aligns with **SDG 9: Industry, Innovation, and Infrastructure** and **SDG 11: Sustainable Cities and Communities**.

Reason for Mapping:

SDG 9: Industry, Innovation, and Infrastructure

1. Promotes innovation by integrating embedded systems, machine learning, and gesture recognition
2. Encourages development of smart and intelligent automation systems
3. Supports advancement in modern home automation technologies

SDG 11: Sustainable Cities and Communities

1. Enables smart home solutions for efficient and convenient living
2. Reduces dependency on multiple physical remote devices
3. Contributes to the development of energy-efficient and connected living environments

ANNEXURE I

BASE PAPER

Article

Gesture Recognition Framework for Teleoperation of Infrared (IR) Consumer Devices Using a Novel pFMG Soft Armband

Sam Young, Hao Zhou and Gursel Alici * 

Applied Mechatronics and Biomedical Engineering Research (AMBER) Group, School of Mechanical, Materials, Mechatronic and Biomedical Engineering, Faculty of Engineering and Information Sciences, University of Wollongong, Wollongong, NSW 2522, Australia; samalexyoung@outlook.com (S.Y.); hzhou@uow.edu.au (H.Z.)

* Correspondence: gursel@uow.edu.au

Abstract: Wearable technologies represent a significant advancement in facilitating communication between humans and machines. Powered by artificial intelligence (AI), human gestures detected by wearable sensors can provide people with seamless interaction with physical, digital, and mixed environments. In this paper, the foundations of a gesture-recognition framework for the teleoperation of infrared consumer electronics are established. This framework is based on force myography data of the upper forearm, acquired from a prototype novel soft pressure-based force myography (pFMG) armband. Here, the sub-processes of the framework are detailed, including the acquisition of infrared and force myography data; pre-processing; feature construction/selection; classifier selection; post-processing; and interfacing/actuation. The gesture recognition system is evaluated using 12 subjects' force myography data obtained whilst performing five classes of gestures. Our results demonstrate an inter-session and inter-trial gesture average recognition accuracy of approximately 92.2% and 88.9%, respectively. The gesture recognition framework was successfully able to teleoperate several infrared consumer electronics as a wearable, safe and affordable human–machine interface system. The contribution of this study centres around proposing and demonstrating a user-centred design methodology to allow direct human–machine interaction and interface for applications where humans and devices are in the same loop or coexist, as typified between users and infrared-communicating devices in this study.



Citation: Young, S.; Zhou, H.; Alici, G. Gesture Recognition Framework for Teleoperation of Infrared (IR) Consumer Devices Using a Novel

Keywords: gesture recognition; teleoperation; force myography; pneumatic myography; armband; machine learning; infrared

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1. Introduction

The modern household in Australia contains, on average, over 19 connected devices and is forecasted to have more than 30 within the decade [1], many of which have a means of interaction/communication with users. These devices are diverse in their range of purpose, functionality, ergonomic attributes and technological advancement. Similarly, the users of these devices are diverse in their communication methods, physical and cognitive abilities, desired intents and technological capabilities. Considering the amount of existing technology and the variance in their production methods, it has become evident that a backward functionality solution is needed—a solution that can map the unique needs of the individual to the unique functions of existing devices.

Current research trends show a growing interest in utilising gesture recognition as a means of human–machine interaction, redefining the concept of user interfaces with consumer electronics of different sizes and capabilities [2]. Gestures are expressive, purposeful body motions that utilise the physical manipulation of a person's hands, arms, face and body to interact with their surrounding environment or to communicate information. Gesture recognition has thus become highly regarded as a reasonable and efficient means of human–computer interfacing [3].

Hand/arm gesture recognition is utilised in a diverse range of topics including prosthesis control, human–computer interaction systems, sign language recognition, musical creation, robotics control and entertainment/virtual reality interfaces [4–8]. The methods employed to recognise gestures include motion tracking, computer vision, electromyography (EMG) [9], force myography (FMG) [10,11] and mechanomyography (MMG) [12]. This work uses a custom-built pressure-based force myography (pFMG) method to acquire user intent [13].

Although optical tracking systems, both marker-based and markerless, are often preferred over EMG and FMG for gesture recognition due to their precision and robustness, these systems have notable drawbacks that make them less practical for everyday use. Marker-based tracking requires the placement of physical markers on the user's body, which can be cumbersome, intrusive and uncomfortable, especially for prolonged use. Markerless systems often suffer from occlusion issues with user's gestures blocked from the camera's view, leading to unreliable recognition. Additionally, markerless systems can be computationally intensive, requiring substantial processing power and sophisticated algorithms to achieve real-time performance, which can be a barrier to widespread adoption.

In contrast, pFMG can offer a more practical and user-friendly approach to gesture recognition, suitable for real-world use. The pFMG sensors used in this paper can detect changes in muscle pressure, providing a direct and reliable measure of muscle activity without the need for visual tracking. This eliminates the occlusion problems inherent in optical systems and reduces the computational load, facilitating faster and more efficient real-time processing. Furthermore, pFMG sensors are non-intrusive and can be comfortably worn for extended periods, making them suitable for continuous use.

This body of work is an extension and application of our previous work developing a soft pressure-based force myography armband [13]. In the previous research, a novel non-invasive, lightweight and low-cost wearable armband was developed for human–machine interface applications. This soft pFMG armband uses air pressure sensors to capture mechanical biosignals from muscle activity, providing stable, reliable, and reproducible data with minimal calibration. The armband's design ensures durability and resistance to external factors like sweat and mechanical deformation. This work extends its capabilities by providing the “intelligence” to the armband—adding a gesture recognition framework and teleoperation control capabilities.

To demonstrate the practical impact and provide a foundation for future research, empirical evidence of the proposed system's real-world relevance and market potential is presented by integrating teleoperation capabilities for infrared (IR) communicating devices. By focusing on IR, it is highlighted how the proposed system can enhance user experience and interaction with everyday devices. The abundance of IR communicating consumer electronics with its wide range of functionality achievable across devices makes it an excellent medium for control. IR ports are common in televisions, DVD players, sound systems, air conditioners, drones, robots, toys and many other household peripherals. IR technology uses the infrared component of the electromagnetic spectrum (870–950 nm), requires a direct line of sight and cannot pass through walls, which can provide a level of privacy and security not achievable by other wireless communication protocols [14]. These factors make IR an effective protocol for the purposes of this study, though future work could benefit from incorporating additional wireless communication methods to supplement the range of targetable devices and leverage their transmission features.

Furthermore, IR devices have been common in households for many years, and are familiar/intuitive to control for all age groups. The combination of multiple devices being controlled by different remotes with numerous buttons can result in households with an overwhelming amount of functionality [15]. With this considered, it is easy to understand how users with cognitive or physical impairments can be disadvantaged when navigating these controls. Ergonomics studies reveal that the majority of users only engage with the direct action controls of their remote devices (i.e., power, play, volume, channel buttons, etc.) [16]. A study [17] reported that remote control usability is not only defined by the

traditional work-related definition but also “emotional usability”, which includes additional needs such as enjoyment and entertainment which enhance the users’ product experience. Our work integrates the benefits above of pFMG and IR technology with a user-centred design methodology to enable direct human–machine interaction with infrared-communicating devices through a novel, pressure-based soft robotic armband. This armband captures user intent and teleoperates targeted IR devices by recognising five different hand gestures (sufficient for action controls) using machine learning algorithms. The key contributions of this paper are:

- This is the first study to investigate and demonstrate the feasibility of using an FMG-based gesture recognition system for teleoperating consumer IR devices.
- Two implementations of the gesture recognition system, the original framework and the extended framework, are proposed and implemented. All phases of the system architecture are thoroughly detailed and analyzed, resulting in a modular solution that adapts to individual users and various devices. Additionally, it is designed to be sufficiently fast, enabling ‘plug and play’ capabilities.
- The selection of feature sets and classifiers are comprehensively studied to identify the optimal configurations for pFMG-based hand gesture recognition.
- Our proposed system demonstrates a state-of-the-art performance in both recognition accuracy and prediction speed when compared to the existing literature incorporating FMG- or EMG-based gesture recognition systems with similar gesture classes.
- This research provides empirical evidence of the system’s practical impact and market potential, emphasising its real-world relevance and facilitating reproducibility for further optimisation. The scientific contribution of this work is especially beneficial to people with upper limb loss/deficiency, allowing them to teleoperate household devices without the need for an intermediary prosthetic limb. The device design and recognition implementation do not require specific armband orientation or placement, as it maps overall muscle intent rather than targeting specific muscles.

2. Materials and Methods

This section presents the equipment, the proposed original framework and extended framework methods to teleoperate various IR consumer devices by hand gestures. The main equipment includes a custom-built pFMG armband and an IR emulator device. In the original framework, all the steps are described in detail, including IR and pFMG data acquisition, pre-processing, feature construction and selection, classifier model selection, gesture mapping, post-processing, and device actuation. After adding a motion sensor (IMU), an extended framework method is proposed to further enhance the functionality of the system with spatial augmentation, quasi-dynamic gestures, and proportional control of a linked device.

2.1. Equipment

A novel pFMG armband prototype (shown in Figure 1) is developed for FMG data acquisition. The armband is made up of five 3D-printed pressure-sensitive chambers (PSCs) that are soft and flexible using Ninjabflex as one of the commercially available flexible filaments. Physical expansion or contraction of the forearm muscles during a gesture execution deforms the chamber which in turn generates internal pressure changes that can be detected by the attached amplified pressure sensors (Honeywell ABPDANT015PGAA5). Our previous work [13] demonstrated that five chambers were sufficient to fully encompass the forearm tightly whilst keeping desirable design features such as modularity (add/subtract sensing chambers) and scalability (increase/decrease diameter) for individual users.

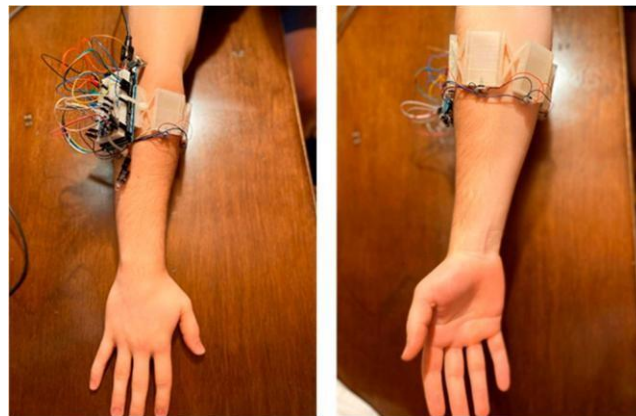


Figure 1. Novel pneumatic myography (PMG) armband [13].

The mechanical characteristics of the Pressure Sensitive Chambers (PSCs) were proven to be linear, stable over time, show insignificant hysteresis and attain a high lifetime cycle (greater than 1.5 million cycles to failure). The PSC's insensitivity to drift means that there is minimal need for re-calibration, which is a primary advantage of this pressure sensing system over existing piezoresistive force/strain sensors on the market. The armband serves as a low-cost (approximately AUD 129 for all the materials) and effective alternative to traditional EMG and FMG sensors. The armband is user-friendly, fully customisable and stretchable, increasing comfortability as it can be worn over clothes without performance effects occurring. The comprehensive details of the armband are introduced in [13].

A low-cost infrared emulator device is developed by using off-the-shelf components and incorporated with the pFMG armband using an Arduino Mega2560 (Arduino LLC, Turin, Italy). An infrared receiver module was placed towards the back of the device to record incoming infrared codes transmitted from existing remotes. An infrared transmitter module was placed at the front of the device to transmit the stored infrared codes back to the peripheral devices when a gesture is performed. A 9-DOF (degrees of freedom) IMU was placed centrally forward on the remote and has capabilities to capture 9 distinct types of orientation- or motion-related data: 3 DOF each of angular velocity, acceleration and magnetic orientation.

An infrared library by Ken Shirriff [18] was used for decoding and transmitting the received IR signals. Finally, machine learning tools from the Scikit-learn software package [19] in Python (v3.7.1) were used. All processing was performed on a laptop with Intel® Core™ i7-8550U CPU (Intel, Santa Clara, CA, USA) and 8GB of installed RAM. It is recognised that this is a prototype design and there are easily a few components that can be replaced with smaller, more efficient and/or customised alternatives. The final device, including the armband and the IR module, weighs approximately 124 g and costs approximately AUD 210 in its prototype form.

2.2. Framework Method

The system framework has been broken down into distinct phases, both in terms of reporting and the system architecture. These phases are as follows:

- Infrared data acquisition—Infrared signals are recorded from their target devices and stored for future emulation.
- pFMG data acquisition—Raw force myography signals are recorded from the forearm muscles of the user when performing gestures.
- Pre-processing—The raw pFMG data are filtered and rectified to establish a form that is suitable for featurisation and ultimately be fed into a classifier model.
- Feature construction and selection—The processed data are formatted and fed into mathematical operations that help differentiate and diverge gesture data.

- Classifier model selection—Classifier models learn how to predict gestures based on training and test datasets from the featured data.
- Gesture mapping—Gestures are mapped to infrared device functionalities.
- Post-processing—Predictions from classifier models are collected and refined to improve the stability and accuracy of final predictions.
- Device actuation—Final predictions are used to control devices according to gesture mapping setup.

The benefit of this modular nature is that entire system capabilities can be modified relatively easily whilst maintaining a functioning workflow. Figure 2 presents a simplified overview of the framework process in the form of a visual process flowchart.

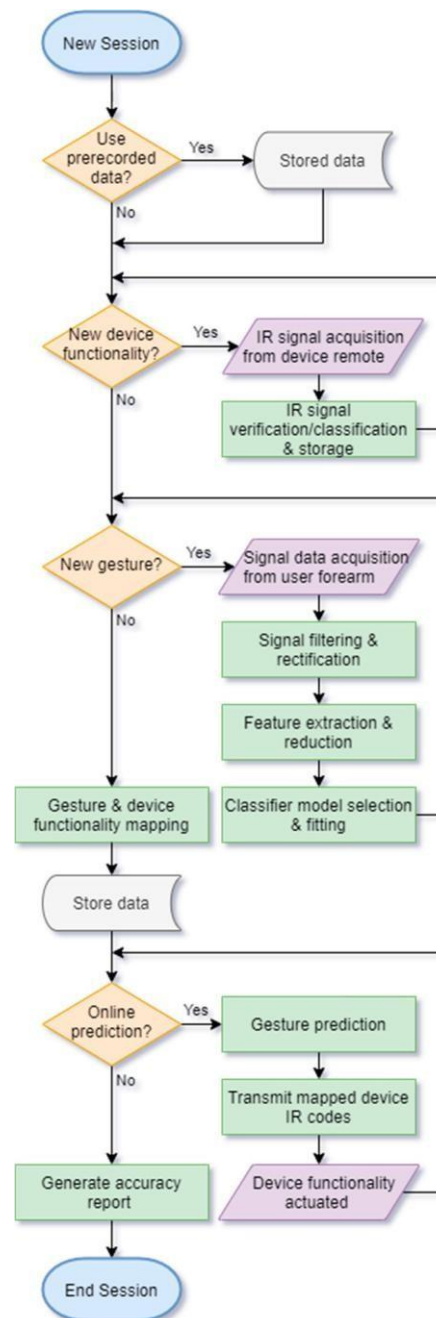


Figure 2. Gesture recognition and device teleoperation framework.

2.2.1. Infrared Data Acquisition

IR codes were recorded by actuating desired buttons of existing IR remotes towards the direction of the infrared receiver. A direct line of sight was maintained to ensure the IR codes were received correctly, i.e., not intercepted, scattered or reflected by different mediums and surfaces. An Arduino script, together with a popular IR remote library by Ken Shirriff [18] decoded the incoming signals based on the protocol pattern received. Our framework has accounted for 17 unique manufacturer protocols and their sub-variations that incorporate a large and varied pool of IR consumer goods (common protocols identified include NEC, Sony and RC5). To account for codes that are not predefined as a particular manufacturer type, our framework simply records the raw code data. This data information is in the form of pulse bursts and space timing ticks which can be stored and replicated in its raw form for future use. This method could have been used for all received IR signals but the former method allows for optimisation and easier identification.

2.2.2. pFMG Data Acquisition

As shown in Figure 3, five common gestures (Fist, Pinch, Spread Fingers, Wave In and Wave Out) were selected to test the effectiveness of the gesture recognition system. These gestures were selected for a number of reasons, the main one being that they are relatively prevalent in the myography-based gesture recognition literature. These gestures are frequently cited as possessing the following desirable characteristics [20–22]:

1. They are discernible enough to be detected by most types of sensors, EMG, FMG, MMG, etc.
2. They are distinguished from each other, i.e., use differing muscle groups during actuation and are sufficiently different from resting position.
3. They are simple to understand and perform. It is recognised that patients with arm/hand disabilities/amputations may still find it difficult or impossible to perform the gestures; however, user intention is relatively clear when performing these gestures (which can still be detected by myography-based systems).
4. They are comfortable to perform, i.e., reduce induced muscle pain and fatigue.
5. They are intuitive to the user in performing operations and/or actuating devices.

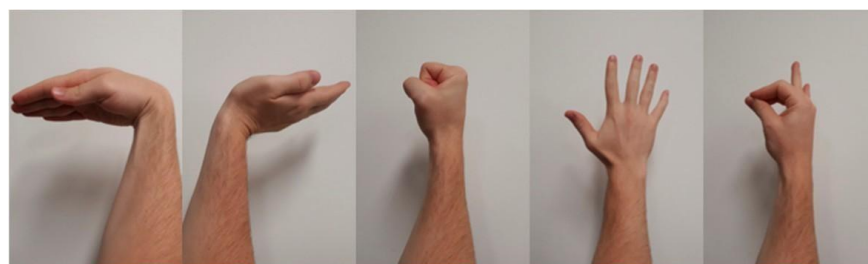


Figure 3. Gestures performed by subjects—(left to right) Wave In, Wave Out, Fist, Spread Fingers and Pinch.

It should be noted, however, that a large body of the published literature utilises EMG-based acquisition devices and thus does not truly reflect the optimal gestures that could be used for the FMG-based method utilised in this study. Finding the optimal gestures for FMG applications was deemed beyond the scope of this study and the disadvantages of using EMG-refined gestures were outweighed by the need for the literature comparability.

For the actual implementation of gesture data acquisition, the pFMG armband was placed over the belly of the forearm (slightly distal to the elbow). The orientation of the armband was adjusted to ensure the base was situated slightly medial to the anconeus muscle. The base contains no pressure chambers and was placed on this region as it has little muscle deformation relative to other sites. Proper orientation was not crucial for gesture recognition but the same orientation between sessions was required unless a user chose to define new gestures or redefine existing ones.

Raw FMG data were recorded by the pFMG armband as a stream of arrays at a sampling period of 1.35 ms (740 Hz), each containing five elements (for each pneumatic chamber). For training the machine learning algorithms, a single gesture recording was defined as obtaining 2000 pFMG data samples which equated to approximately 2.7 s of data. This amount of data was deemed sufficiently long to encompass a static pose whilst not causing fatigue effects during training sessions when the user would have to repeat gestures multiple times in a row. The number of times a recording trial (each of the 5 gestures performed for 2.7 s) was repeated was set to 3. Utilising 3 separate recordings of a 3-second gesture (with other gestures interspersed) instead of a single 9-second gesture recording increased the variability in subject data, preventing the likelihood of overfitting and reducing muscle fatigue. Users invariably made discrete adjustments to their actuation of a gesture upon re-initiation.

2.2.3. Pre-Processing

In preparation for featurisation, the gesture data were segmented by defining a ‘window’ of 100 samples (approximately 135 ms worth of data) and stepping through the data with a stride length of 50 samples (allowing for 50% overlap) as depicted in Figure 4. Sliding window implementation with 50% overlap (not to scale). For a given gesture recording length, this would produce $(N/S - 1)$ windows, where N is the number of samples and S is the stride length. Due to the consistent, steady nature of the pressure and orientation readings, it was identified that little rectification or smoothening was required prior to featurisation. As a precautionary measure, a 5th-order digital bandpass filter was utilised to limit a 10% sampling rate margin of error. This would ensure that any lag or lead effects would be disregarded and alert the user of invalid data. No such alerts were recorded during the process of experimentation.



Figure 4. Sliding window implementation with 50% overlap (not to scale).

2.2.4. Feature Construction and Selection

Featurisation is the process of transforming raw FMG signals obtained from the armband into features that better represent the data to classifier models. This increases the predictive capabilities of the machine learning models and improves their accuracy when subjected to unseen data. In this study, this featurisation is achieved through the application of mathematical operations to each window of gesture data. A total of 17 time-domain features were initially applied including Integrated Absolute Value (IAV), Waveform Length (WL), Average Amplitude Change (AAC), Slope Sign Change (SSC), Simple Square Integral (SSI), Root Mean Square (RMS), Mean Absolute Value (MAV), Modified Mean Absolute Value Type 1 (MAV1), Modified Mean Absolute Value Type 2 (MAV2), Zero Crossing (ZC), 3rd Temporal Moment (TM3), 4th Temporal Moment (TM4), 5th Temporal Moment (TM5), Willison Amplitude (WAMP), Difference Absolute Standard Deviation Value (DASDV) and Myopulse Percentage Rate (MYOP) (see [23–28] for mathematical descriptions and operations). The final list of the features selected for the purpose of this study is presented in Section 3.4.

Feature construction transforms a single window of gesture data into a ‘featured window’ which is a comprehensive mathematical description of approximately 135 ms of data from a single sensor. A ‘featured frame’ is an amalgamation of each sensor’s featured windows. Therefore, a single gesture can be effectively defined as an array of frames. Intra-gesture variation in features was determined by subjecting element-wise relative standard deviation (coefficient of variation) tests. Inter-gesture variation in features was obtained by applying coefficient of variation tests on the intra-gesture means (mean values between the frames of all compiled gestures).

Feature selection was based on the results of the intra-gesture and inter-gesture variances. A feature was considered highly effective if it was characterised by both low intra-gesture variability and high inter-gesture variability. This demonstrated that the gesture was consistent during its execution and/or between trials but sufficiently different from other defined gestures to be easily distinguished. This method of feature selection follows a filter-based approach, that is, it chooses the optimal features without involving the final classifiers in the process (as is the case for wrapper-based methods). This method prevents the influence of overfitting errors affecting feature performance. Furthermore, a filter-based approach has a significant reduction in processing time when compared to wrapper-based approaches because the latter must iterate the selection process with every combination of feature and classifier model (an exponential function). Following feature selection, the data are formatted into a dimensionality that is suitable for feeding into the classifier models.

2.2.5. Classifier Model Selection

Similar to the protocol used in our previous work [29], the test pool of classifier models started with Decision Tree, Random Forest, k-Nearest Neighbour, Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), Nu-Support Vector Classification (NuSVC), and Gaussian Naïve Bayes (refer to [30,31] for further details). These classifier models are well developed in the existing literature and have substantive documentation for their implementation.

For model evaluation purposes, the trial datasets were split into training and testing sets by the ratio 2:1. Therefore, for every three trials recorded, one trial was hidden from the classifier for future testing purposes. Every unique combination of hidden trials was used to determine the overall classifier accuracy for the given data set. Therefore, the overall classifier accuracy is based on the average of the binomial coefficient accuracy scores according to the binomial coefficient equation:

$$C(n, r) = n! / (r!(n - r)!), \text{ for } r \text{ trials from a set of } n \text{ trials} \quad (1)$$

As a total of 6 trials were recorded per user in the experimental procedure, the classifier accuracy was taken as the average of 15 unique accuracy scores. This method of estimating the skill of the classifier models is essentially a customised version of a k-fold cross-validation method. The primary difference between the two is that due to the relatively small data collection size and the use of static gestures, complete segregation of the trials was necessary to avoid overfitting.

2.2.6. Post-Processing

Due to the high-frequency sampling rate and small data window sizes, the classifiers were able to produce approximately 10 gesture predictions per second. Post-processing was performed on the system to condense these predictions into a single prediction which increases the stability and accuracy of the results. A prediction queue was established and after it was filled to a maximum of 10 predictions, the mode value was taken as the final prediction. Note that if the mode value did not constitute at least 75% of the queue, the predictions were deemed too random, the gesture was labelled as 'Unknown' and no IR codes were transmitted. If desired, the last confident prediction can be used as the current gesture during these 'Unknown' gesture periods. This is useful for scenarios where the user is experiencing fatigue as a result of transmitting repeat codes over an extended period of time.

2.2.7. Gesture Mapping to IR Codes

After both gestures and IR codes were recorded, the user was given the option to map gestures to any number of IR codes. Assigning the same gesture to multiple codes allows for the teleoperation of multiple devices using the same gesture. This is beneficial when a gesture is intuitive for multiple cases, for example, an outwards wave gesture could

allow for both turning up the volume on a speaker and temperature on an air-conditioner. Mapping multiple IR codes to the same gesture also allows for synchronous teleoperation of multiple devices. This technique is, however, limited by the line-of-sight transmission angle that is a characteristic of infrared communication. Alternatively, the user could map multiple gestures to the same infrared code, though this would only serve to provide added redundancy in the system as a backup in the case of poor gesture recognition.

2.2.8. Device Interfacing and Actuation

The current framework's device interfacing capabilities were tested on a variety of infrared communicating devices. To demonstrate the efficacy of the system, a number of factors were taken into consideration when selecting devices to actuate:

1. The devices must be prevalent in society, i.e., commonly used by people in a variety of environments including homes, offices and factories, etc.
2. The devices must demonstrate a variety of consumer electronics, i.e., showcase actuation of television units, air-conditioning units, sound systems and video players, etc.
3. The devices should utilise a variety of manufacturer protocol codes with different modulation methods, i.e., NEC, RC5, Sony and Mitsubishi, etc. using 12- and 16-bit protocols.

IR codes were recorded onto the IR transceiver by triggering commands from IR actuators (e.g., TV remote). These recorded codes (unique to a given manufacturer, device and button) were subsequently transmitted back to the controlled end device (e.g., TV) when the mapped gesture had been activated. Note that IR codes were only transmitted when a gesture was changing state, that is, repeat recognition of the same gesture did not result in repeat transmission of the IR codes as this prevented debouncing-characteristic behaviour. The current gesture and IR code transmissions were observed by the user in real-time in text form within the application package.

2.3. Extended Framework Method

2.3.1. Spatial Augmentation

The extended framework follows the same general phases as the original framework (Section 2.2) with the exception of spatial augmentation added to the actuation capabilities. This was primarily carried out as a proof of concept with the goal of achieving quasi-dynamic gesture recognition functionality. This framework was intended to be an extension and thus the aim was not to alter the foundation of the original framework. This would mean that the augmentation capabilities could be turned on and off seamlessly in real-time as desired without affecting performance.

Spatial data were added as part of the extended framework by incorporating a 9-DOF motion sensor, IMU, (BNO055, Bosch Sensortec, Reutlingen, Germany) in the pFMG armband. Although the sensor had an in-built accelerometer, gyroscope and magnetometer, these features were omitted as the framework was only concerned with the absolute orientation of the forearm. Modules of the framework do allow for future expansion, incorporating information such as angular velocity and acceleration vectors. Acceleration and velocity data acquisition was found to require a more rigorous calibration setup, sometimes taking as long as 30 s to ensure a sufficiently high confidence level. The nature of this framework was to promote fast and easy utilisation by end users and so it was decided to neglect this information for this version of work and utilise Euler vectors as a means of spatial augmentation. Sampled at a rate consistent with the original framework (1.35 ms), an Euler vector provides absolute orientation information in three axes (x, y, and z).

Initially, the orientation data were included within the featurisation process; however, this immediately presented a significant issue. Defining gestures with integrated orientation values meant that every new orientation was defined as a different gesture. This made the method impractical as there were infinite orientations in 3D space. A potential fix would be to place a sensor onto the torso of the user to give an initial frame of reference and allow

for relative orientation data. This, however, poses issues in itself; the extra sensor increases the bulkiness of the final devices, requires an attachment mechanism, increases the overall cost, and technical complexity and furthermore the user's torso would have to remain still.

The alternative solution was to remove the orientation data from within the feature and classification process and instead use it within the post-processing phase of the framework. This meant that orientation data were only considered in a complementary form after a gesture had been predicted. This had three primary benefits:

1. As a complementary approach, no significant modifications to the original framework were necessary.
2. Spatial gestures were now relative to the starting position of the gesture (as opposed to the infinite starting positions relative to the 3D orientation space).
3. Gestures were now more stable across different orientations. For example, prior to spatial augmentation, slight muscle deviations caused by changing the orientation while holding a static gesture may have influenced the prediction of a gesture. With spatial augmentation, the gestures were more 'locked in'—i.e., it was recognised that the user had moved whilst holding a gesture (less likely to be perceived as a new gesture altogether).

2.3.2. Quasi-Dynamic Gestures

Spatial augmentation allows for quasi-dynamic gestures to be recognised in the extended framework. A quasi-dynamic gesture differs from a dynamic gesture in that the former actuates a constant gesture (no transient motion) with variable orientation and the latter actuates a transformative gesture (with transient motion). There are a few reasons as to why quasi-dynamic gestures were implemented instead of dynamic gestures in this study. One of the key components of this system is its ability to perform FMG data acquisition, pre-processing, featurisation and classification prediction in a matter of milliseconds (approximately 10 predictions can be made per second). This speed is possible as a result of the real-time sliding window implementation with small packets of data, essentially taking many snapshots of the data to understand discrepancies in motion. This implementation is not as well suited to dynamic control as it is for quasi-dynamic control. Dynamic control requires a lagging window and holding off on final predictions to ensure a gesture is completed first. Without any modifications, the original framework would potentially identify the dynamic wave gesture in the top row of Figure 5 as a series of discrete gestures, quickly transitioning between Wave In, Rest and Wave Out (with perhaps 'Unknown' in-between states). Another consideration of dynamic gestures would be the temporal parameter of movement speed. There are virtually an infinite amount of speeds that the gesture could be actuated with and thus the movement will have to be scaled in real-time (with dynamic time warping algorithms) to measure similarities between the trained and recorded temporal sequences.

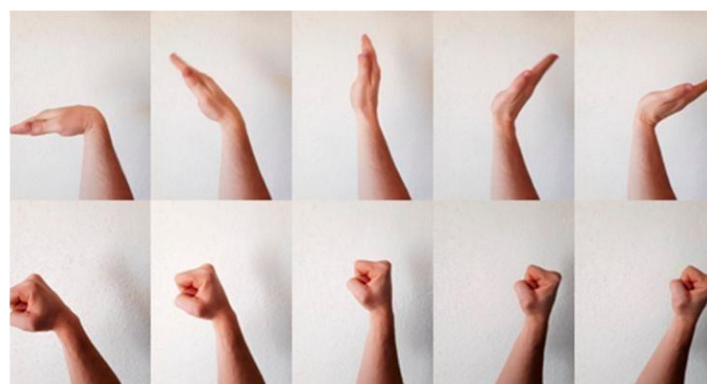


Figure 5. Comparison between dynamic gestures (**Top row**) and quasi-dynamic gestures (**Bottom row**).

Quasi-dynamic gestures do not require any scaling as the gesture relative to the arm-band is constant. This significantly reduces the processing power required and processing time until a final prediction as no lagging is necessary. The quasi-dynamic gesture in the bottom row of Figure 5 is still considered to be a static Fist gesture in the eyes of the classifiers but the orientation change allows for additional functionalities to be added during the post-processing stage.

2.3.3. Proportional Control

Proportional control is a linear feedback control system in which a correction is applied to a given variable with a magnitude that is proportional to the difference between a desired and recorded value. In this extended framework, proportional control was possible by changing the orientation of the arm whilst a gesture was being held. The setpoint orientation was established during the initiation of the gesture and subsequent movements determined the process variable. The difference between the instantaneous orientation values of the initial and current poses influenced the magnitude of the controller output in a linear fashion. Depending on the gesture being performed, only one axis was considered (though multiple axes could be considered independently). To clarify with an example, a 'Fist' gesture that controls an LED strip was activated in an anterior orientation before the arm was laterally rotated. In this example, the system was designed to compute the difference in the orientation of the longitudinal (craniocaudal) axis. The magnitude of lateral rotation from the starting setpoint was then subsequently used to set the brightness of the LED strip. As the infrared codes were transmitted in discrete packets as opposed to a continuous stream, the proportional control was used to either determine the rate of sending the same codes (e.g., increase television volume faster when laterally rotated) or incrementally select the next stored infrared code (increase television channel number when laterally rotated).

2.3.4. Extended Framework Gesture Dataset

To the best of our knowledge, there are no established gesture datasets for the accuracy comparison of quasi-dynamic gesture recognition systems. This study therefore determined to test the augmented version of the five common gestures used in the original framework. These quasi-dynamic gestures were as follows: Fist with supination/pronation rotation; Pinch with medial/lateral rotation; Spread Fingers with arm flexion/extension; Wave In with flexion/extension and Wave Out with medial/lateral rotation. These covered rotations about the longitudinal (craniocaudal), anteroposterior (dorsoventral) and horizontal frontal (mediolateral) axes.

Using the same procedure as the original framework, the raw FMG data were recorded from the upper forearm by the pFMG armband at a sampling period of 1.35 ms. In total, 2000 FMG samples (2.7 s of data) were obtained per gesture for a single trial with 3 trials recorded in total for a user session. The only adjustment was that users were asked to perform the orientation motions slowly during the 2.7 s recording time, taking care to keep the gesture consistent and ensure the pressure chambers were not pressed against the body during rotation.

3. Results

This section presents the hand gesture recognition performance using the pFMG armband and the proposed framework methods. Some better-performing features and classifiers are identified. The offline classification results show comparable gesture recognition accuracy to some previously reported work. Furthermore, the feasibility of online teleoperation of various IR devices by the proposed system is successfully demonstrated.

3.1. Participant Subjects

The gesture recognition accuracy component of the original framework was evaluated by utilising 12 able-bodied subjects (9 males, 3 females, age 23.1 ± 2.2 years; mean $\pm$ SD). The experiment methodology was approved by the Human Medical Research Ethics Committee at the University of Wollongong and each participant signed consent

forms for their data to be included within this report. The inclusion and exclusion criteria for participant recruitment are described below.

Participants must meet the following:

- Be 18 years of age or older.
- Have no known musculoskeletal or neurological disorders affecting the upper limbs.
- Be able to understand and follow instructions in English.

Participants will be excluded from this study if they meet the following:

- Have skin conditions such as dermatitis or hypersensitivity that could be aggravated by contact with electrodes or wearable devices.
- Have epilepsy or a history of seizures, as the experiment may involve visual or auditory stimuli that could trigger an episode.
- Are pregnant, as the effects of some sensory modalities on pregnancy are not well studied.
- Have recently undergone surgery on their hands, wrists, or forearms.

pFMG data from the forearm muscles of their dominant hand were recorded using the pFMG armband. Each participant was asked to wear the pFMG armband on the belly of their dominant hand's forearm, positioning the base chamber (which contains no sensors) slightly medial to the anconeus. If the armband was deemed to have insufficient contact pressure (perhaps due to a smaller forearm circumference), then the base chamber was detached to reduce the armband circumference. For the offline analysis of gesture recognition using the original framework, 5-channel pFMG data were recorded in two sessions for each participant without any reported effects of fatigue. In each session, a single trial consisted of performing the five gestures (Wave In, Wave Out, Spread Fingers, Fist and Pinch) in a random sequence for approximately 3 s each. The trial would then be repeated twice more for a total of three trials per session, each gesture containing approximately 9 s worth of data. The entire session would then be repeated, resulting in 18 s worth of data per gesture.

For the extended framework, only 3 participant subjects (1 male, 2 females, age 32.3 ± 13.2 years; mean $\pm$ SD) were available for testing. This was deemed insufficient for robust conclusive results but proceeded nonetheless as a proof of concept and for experimentation purposes.

3.2. Infrared Emulator Results

The infrared codes of all devices tested were able to be received, stored, emulated and transmitted by the infrared emulator device. It was noted that manufacturer codes differ in the number of bits used (e.g., NEC uses 32 bits while Sony uses 12 or 20 bits). Some manufacturer codes are sent numerous times for each button press and a few remotes were noted to use more than one type of manufacturer code. Additionally, holding down a button transmits repeated codes for some remotes whilst for others it produces a different, symbolic code altogether. The current method of overcoming these issues is to manually specify the above parameters.

While all tested devices were eventually able to be replicated, the most challenging emulations stemmed from unknown manufacturers where the protocol modulation scheme is indeterminate. For these cases, the codes were recorded in their raw form and labelled as 'Unknown' type. This was recorded to be slightly less efficient in processing time when compared to standardised, recognised protocols but was virtually imperceptible to the user. The main issue with accepting recorded 'Unknown' protocols was that there was no way to distinguish between correct protocols from unknown manufacturers and incorrect protocols from a scrambled recording (reflections off surfaces, dissipated through mediums, etc.). Although it was not an ideal solution, the 'Unknown' protocols were recorded multiple times to check for repeated codes. Scrambled codes are virtually random, and therefore any repeated codes were indicative of the real protocol pattern.

3.3. Characterisation of Static Gestures

The characterisation of the static gestures (Wave Out, Wave In, Spread Fingers, Fist and Pinch) are presented in Figure 6. It was observed that the pressure output values from each of the chambers were relatively stable and showed little hysteresis effects. This partially demonstrates why filtering and rectification were not entirely necessary for this data type. The bandpass filter applied contained a function to alert the user if there was ever an abnormal spike in pressure data. The threshold was defined as a 20% change in pressure value over a period of less than a second through the duration of a single static gesture. No alerts were recorded and thus the bandpass filter was deemed not necessary for this data set but allowed for a fail-safe mechanism.

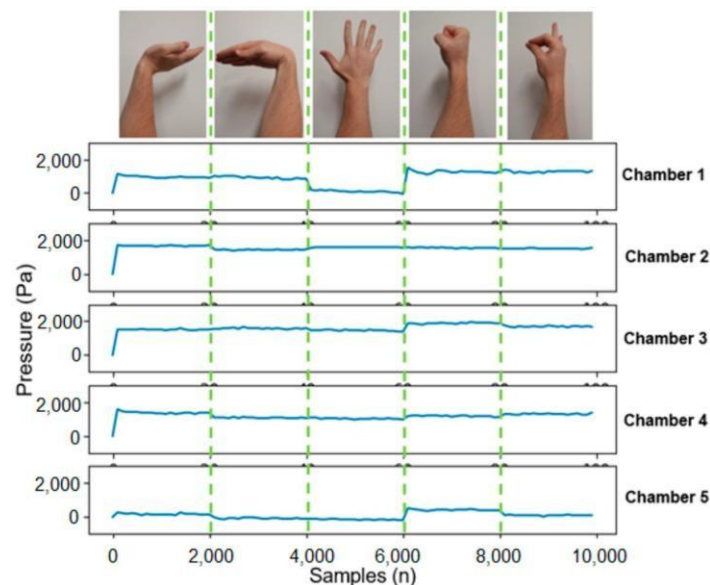


Figure 6. Characterisation of gesture recordings for 1 trial (2.7 s of data per gesture).

A control armband output value was attained by placing the armband on the ground with no disturbances. This determined the base pressure already contained within each of the chambers. It was noted that the base pressure within each of the five chambers differed, but this was considered insignificant due to their small values and because the output pressures for classification are relative to initial user pressure.

Observation of the data proved very useful in identifying optimal chamber area placements to allow for highly distinguished gestures. To clarify, in Figure 6, Chamber 1 had a clear discrepancy between the Spread Fingers gesture and the Fist gesture whereas Chamber 2 had virtually no discrepancy between the gestures. To understand the reasoning behind this drastic change, it must be noted that Chamber 2 sits approximately where the pronator teres muscle lies in the forearm. The Fist and Spread Fingers gestures both do not require movement of the pronator teres and thus remain at relatively the same pressure level. In fact, Chamber 2 remains relatively the same pressure level across all 5 gestures. This does not necessarily indicate that Chamber 2 is a redundant chamber (therefore allowing for similar gesture recognition accuracies with only four chambers); however, it does indicate the lack of variety in the chosen gestures. As stated previously, these five gestures were chosen on the basis of being used extensively in the EMG-based gesture recognition literature. These results indicate that having a gesture that requires pronation of the forearm would increase the variability of Chamber 2 data and thus increase its contribution to gesture definitions.

The waveforms in Figure 6 represent data from a single volunteer and can vary significantly between different individuals. Additionally, when a volunteer doffs and dons the armband between sessions, their own waveforms can differ each time. Consequently, the armband classifier typically needs to be retrained for each recording session to achieve optimal results. In this study, the calibration training time was 12 s per gesture, which was

considered an acceptable trade-off to ensure that armband orientation and placement do not significantly impact the classification process.

3.4. Experimental Results—Feature Selection

Our framework defined optimal features to be those that increased the inter-gesture variance of data and simultaneously decreased the intra-gesture variance of data. This meant that the data of a given gesture was repeatable and consistent within itself whilst still being effectively distinguished from other recorded gestures. Initially, a mask was used to highlight which features produced an intra-gesture variance of $<10\%$ and an inter-gesture variance of $>20\%$. These percentage thresholds were also varied, but ultimately, this method proved to be ineffective as a feature could be deemed effective for one chamber but not another, even within the same gesture. Instead, the thresholds were removed, and features were ranked by the intra-gesture and inter-gesture variance average scores across the chambers. This whole process highlighted a potential alternative for classifier selection in future work. It may be beneficial to have a classifier model allocated to each chamber (in parallel) and then the amalgamation of the predictions is fed into an overall classifier (in series) to decide the final gesture based on constituent predictions.

Based on our ranking method, the top 50% of features were chosen as being the most useful. This included (in order) the following features: RMS, IAV, SSI, VAR, MAV1, MAV2 and MAV3. It was observed that feeding classifier models these features (thereby reducing the feature by half from 14 to 7) resulted in improvements in prediction accuracy (approximately 2–3%). This may seem insignificant, but note that this was for a pre-established effective/customised feature set and thus it demonstrated a significant improvement to standardised feature sets. In fact, the removal of the features altogether resulted in an accuracy drop between 16 and 37%, despite the relative stability in the pre-processed pressure values (it is believed this drop would be even more significant for EMG-based systems due to a relatively higher variation in the EMG signals). This drop ultimately signifies the importance of the feature selection process.

A factor that should be considered in the results obtained is that the effectiveness of these features was tested independently of each other; that is, feature combinations were not analysed. This could result in two or more features being classed as effective—independently they would increase the prediction accuracy but combined would not further increase the prediction accuracy because they are describing similar class differentiations. It is hypothesised that this may be the case for features such as MAV1, MAV2 and MAV3. The ability to distinguish effective features was additionally hindered by the difference in rankings of intra-gesture and inter-gesture variances. Some features were ranked highly in having a large inter-gesture variance (positive attribute), but simultaneously had a low-ranking score due to their large intra-gesture variance (negative attribute), e.g., TM3. When it came to these edge cases, higher scores in the inter-gesture variance rankings were weighted more than intra-gesture variance scores. It was ascertained that two gestures could be significantly varied within themselves and still be sufficiently distanced from each other for easy differentiation.

3.5. Experimental Results—Classifier Model Selection

3.5.1. Inter-Trial and Inter-Session Gesture Recognition

The seven classifier models used in this study were evaluated in their performance with respect to accuracy, log loss and training/prediction times. Firstly, the classifier models were fed for the participant sessions individually to determine the inter-trial gesture recognition results.

The average results of all participants for the inter-trial gesture recognition capabilities are summarised in Table 1. Classifier model inter-trial quantitative results. The LDA classifier model is highlighted as the optimal model due to its accuracy score being the highest recorded on average and with the least amount of standard deviation. The training/prediction time was not the smallest recorded out of all the models, which was achieved by the GaussianNB model. It was decided, however, that accuracy takes prece-

dence as the times are so small in nature that there is an unperceivable difference between the majority of models from the user perspective.

Table 1. Classifier model inter-trial quantitative results.

Classifier Model	Accuracy (%)	Log Loss	Training/Prediction Time (ms)
GaussianNB	77.18 ± 19.95	0.068 ± 0.124	3.9 ± 1.4
NuSVC	85.32 ± 11.02	0.070 ± 0.100	72.6 ± 4.0
QDA	84.11 ± 10.25	0.226 ± 0.333	5.6 ± 1.5
LDA	88.93 ± 8.70	0.009 ± 0.015	8.8 ± 1.2
RF	83.88 ± 8.87	0.004 ± 0.005	98.6 ± 2.1
K-NN	86.43 ± 11.85	0.003 ± 0.005	5.2 ± 0.9
DT	82.14 ± 9.66	0.000 ± 0.000	5.9 ± 1.3

The average results of all participants for the inter-session gesture recognition capabilities are summarised in Table 2. The LDA classifier model was again highlighted as the optimal model due to its accuracy score being the highest recorded on average and with the least amount of standard deviation. Similar to the inter-trial gesture recognition results, the GaussianNB model was recorded to have the smallest training/prediction time but was imperceivably different from the LDA model and thus the latter was chosen as the optimal and final classifier model. A sample of the participant classifier model results with respect to accuracy, log loss and training/prediction time is presented in Figure 7.

Table 2. Classifier model inter-session quantitative results.

Classifier Model	Accuracy (%)	Log Loss	Training/Prediction Time (ms)
GaussianNB	87.75 ± 13.62	0.128 ± 0.097	4.6 ± 0.7
NuSVC	90.68 ± 11.31	0.013 ± 0.009	273.9 ± 3.4
QDA	90.12 ± 11.84	0.105 ± 0.096	6.8 ± 1.2
LDA	92.20 ± 9.35	0.010 ± 0.009	16.1 ± 1.1
RF	88.15 ± 10.57	0.001 ± 0.001	161.3 ± 4.2
K-NN	91.20 ± 11.90	0.001 ± 0.001	7.5 ± 0.9
DT	86.18 ± 11.19	0.000 ± 0.000	15.9 ± 2.3

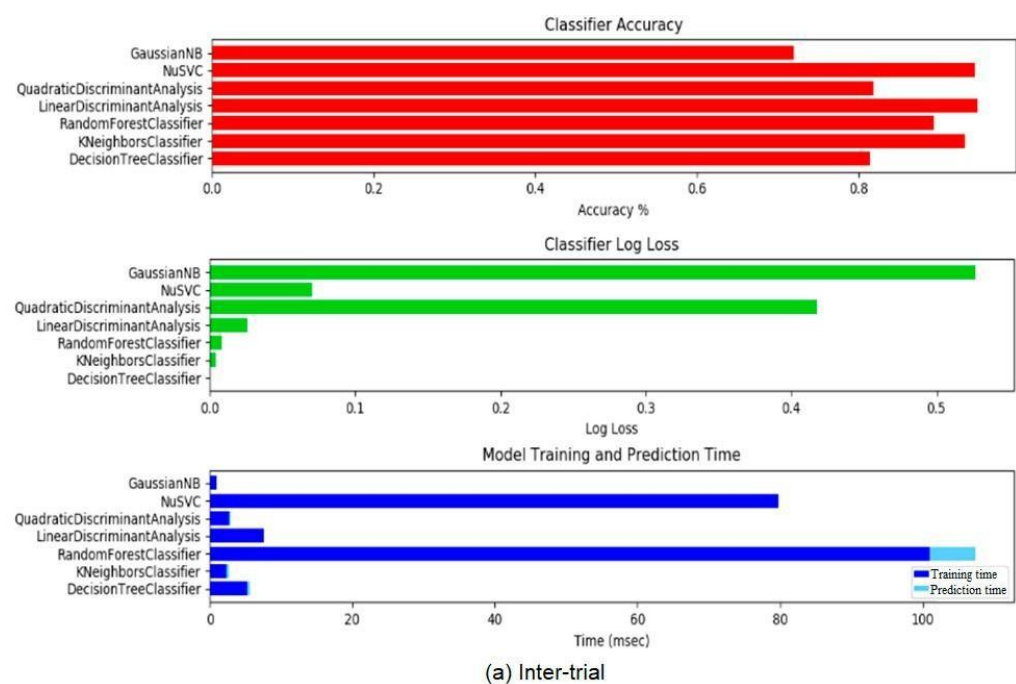


Figure 7. Cont.

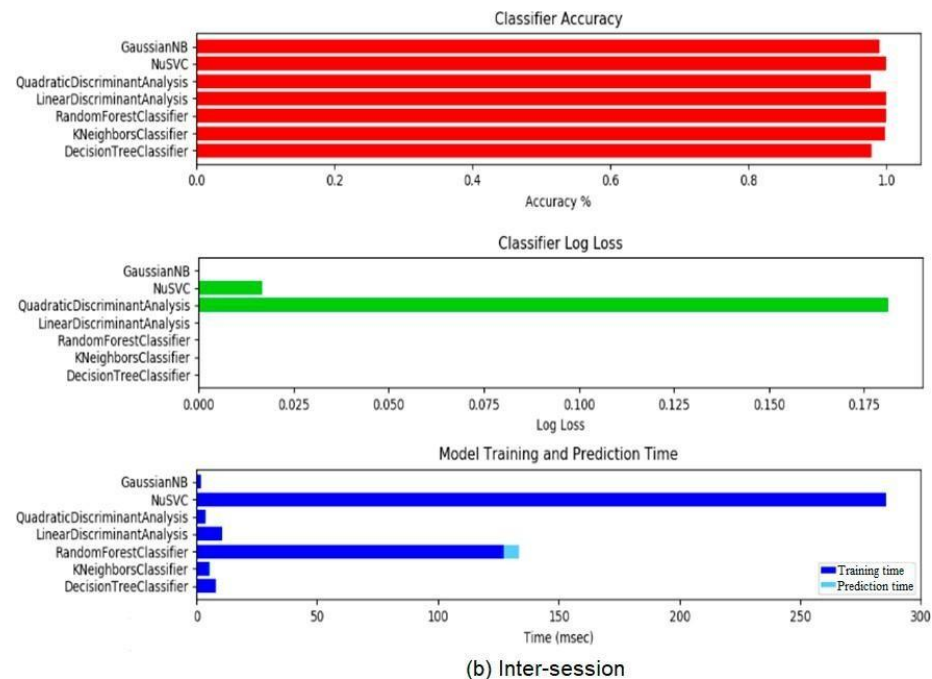


Figure 7. Sample of participant classifier model results with respect to accuracy, log loss and training/prediction time, note scaling ((a) inter-trial; (b) inter-session).

3.5.2. LDA Classifier Model Certainty/Confusion Matrices

Following the selection of the LDA classifier model as the primary machine learning model, normalised confusion matrices (Figure 8) were generated for each participant to visualise the certainty of the predictions. Qualitative observations of these matrices highlighted some interesting conclusions that could be drawn from the data.

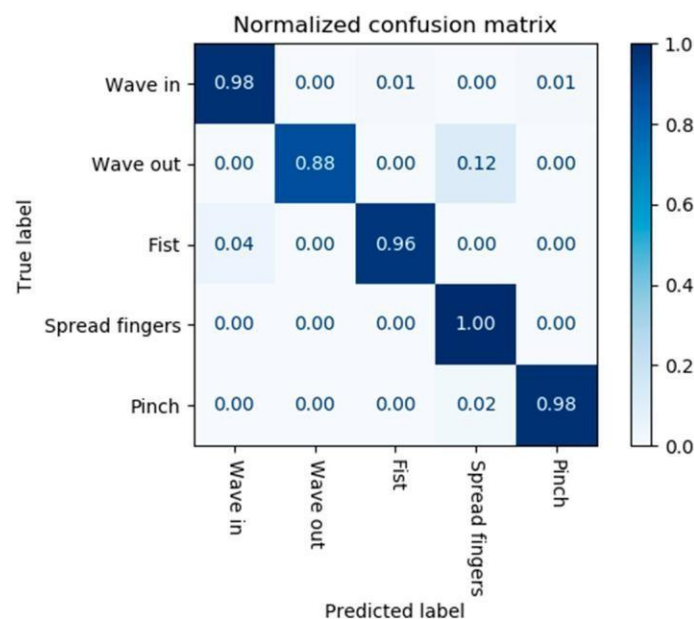


Figure 8. Normalized confusion matrix for given participant sample with LDA classifier model.

There were multiple cases of minor labelling confusions between the 'Wave Out' and 'Spread Fingers' gestures. It is postulated that these minor confusions may be the result of common muscle groups being displaced in the posterior forearm (e.g., extensor digitorum and extensor carpi ulnaris). Furthermore, the 'Wave Out' gesture could understandably be perceived as a continuous form of 'Spread Fingers', i.e., there is no significant change in

finger/arm movements between the two. If a participant is feeling fatigued or bored, their 'Wave Out' gesture may become slackened to look more like a 'Spread Fingers'.

Another interesting feature to note is the difference in participant confusion matrices as a whole. Some participants had exceptionally high scores (97%+) across all gesture labels while those that had poor label predictions tended to over-predict a single gesture instead of an even distribution of incorrect labelling. It was noted that those with exceptional scores tended to have the armband pressed tighter against their forearm. This intuitively makes sense as pressure sensor chambers of the armband are able to be displaced with greater magnitude and be affected more by deep muscles. The tightness of the armband was not formally recorded as a quantitative or qualitative measure as part of the experimental procedure. The prototype armband used in this body of work had two circumference sizes (through the detachment of a non-sensing chamber, which does not include stretching). In an ideal scenario, the armband would be customised through 3D printing techniques to fit the individual's forearm adequately [13].

3.6. Comparison with Existing Literature on Offline Gesture Recognition Results

Comparison of the original framework to the existing literature is made difficult by the fact that none of the literature has performed gesture recognition with a device working on the same principle as the pFMG armband described in this study. Comparisons are further complicated by varying parameters including differing numbers of gestures; differing data acquisition methods and devices; differing machine learning systems; differing recording times per gesture and variabilities in participant pools. The extended framework is even more difficult to compare as it incorporates quasi-dynamic gestures which is an underdeveloped area of research in the gesture recognition literature.

The hand gesture recognition performance by the proposed system was compared to some previously reported works, shown in Table 3. It was decided to set the proprietary Myo armband system (developed by Thalmic Labs [32]) as the benchmark for comparison of gesture recognition systems. Although the Myo armband is an EMG-based acquisition device, it is often used in comparison to FMG-based acquisition devices as they are used in similar applications and have somewhat similar recognition accuracies [26,27]. Furthermore, the Myo has extensively been used in academic and clinical studies [33–36]. Lastly, the five classes of gestures used in the control of the Myo armband are 'Wave In', 'Wave Out', 'Pinch', 'Fist' and 'Spread Fingers'. As these gestures are associated with the Myo armband, they too frequently appear in gesture recognition studies (hence why these gesture classes were utilised in our framework). No such widespread standards in terms of acquisition device or gesture classes could be found for an FMG-based armband and/or framework.

The literature has reported the Myo proprietary system to have an average accuracy score of 83% [33,37]. This score was based on a dataset of EMG recordings obtained from 10 participants, but no recording times per gesture or processing times were included within these reports.

An EMG-based framework [33] claimed to achieve a greater recognition accuracy of 86% using the Myo armband and the same five gesture classes. This framework utilised k-Nearest Neighbour and dynamic time-warping algorithms as part of their machine learning model structure. Their study utilised 10 participant subjects and obtained a training set of 10 s of data per gesture. The gesture recognition system worked in real-time and was able to produce gesture predictions every 250 ms on average.

Another study used both an EMG- and FMG-based framework to determine the gesture recognition accuracy of American Sign Language (ASL) digits 0–9 [38]. Five participants performed ten trials each to assess the system. Each trial contained five seconds of data per gesture and five seconds of rest between gestures. For the EMG-based system, this work achieved a recognition accuracy of 81.5%. For the FMG-based system, a slightly reduced accuracy of 80.6% was obtained. The system had more gestures to classify

compared to the previously mentioned studies, but also contained significantly more data which are factors that need to be considered when assessing the efficacy of these results.

An FMG-based study [10] claims to have achieved an accuracy of 89.4% in cross-trial evaluations for classifying 48 hand gestures. The armband used 16 Force Sensitive Resistors (FSRs) and was long enough to cover both the forearm and wrist. The hand gestures involved 16 sign language gestures, 16 individual finger movements and 16 grasps. The individual recording and processing time per gesture was not clearly specified but it was mentioned that a session consisted of performing each gesture for 5 s and resting for 5 s, repeating 10 times each. This would result in a 40 min long session and a second session was used to further increase the gesture dataset (after approximately 2 h rest).

The performance of the gesture recognition component of our framework was evaluated by taking the average accuracy from the results of 12 participant subjects. As stated in Section 3.5.1, the final classifier model used in the original framework gesture recognition system was the LDA model. To summarise the final results, the gesture recognition system was able to achieve an inter-trial accuracy score of $88.93 \pm 8.70\%$. The inter-trial dataset consisted of approximately 6 s worth of training data and 3 s worth of testing data. The inter-session gesture recognition accuracy, based on training and testing datasets of 12 and 6 s of data per gesture, respectively, was found to achieve an accuracy score of $92.2 \pm 9.35\%$.

It should be noted that the accuracy scores reported are for the individual gesture predictions made by the LDA classifier. In the final implementation of the system for online control of a device, post-processing gathers as many as 10 predictions per second to create a final gesture prediction that is greater in accuracy than its constituent predictions. It is hypothesised that the existing literature may also have implementations of post-processing to increase the final accuracy; however, this is not frequently reported in the literature due to its inability to be compared to others as a singular process.

The primary reason for using the five chosen gesture classes was for easy comparison to the existing literature. Although not quantitatively measured in this study, numerous other gestures were tested (Figure 9) to ensure that a user could effectively use their own custom gestures to control their devices.

Table 3. Summary of similar gesture recognition studies.

Device/Study	Type	Avg. Accuracy	Gestures	Participants	Data per Gesture	Notes
Myo Armband [32]	EMG-based	83%	Wave In, Wave Out, Pinch, Fist, Spread Fingers	10	Not specified	Extensively used as a benchmark; lacks detailed recording times and processing
EMG-based Framework [33]	EMG-based	86%	Wave In, Wave Out, Pinch, Fist, Spread Fingers	10	10 s	Utilises k-Nearest Neighbour and dynamic time-warping algorithms
ASL Digits Recognition [38]	EMG-based	81.50%	ASL Digits 0–9	5	5 s	Performed 10 trials each; uses EMG for recognition
Recognition [38]	FMG-based	80.60%	ASL Digits 0–9 48 hand gestures	5	5 s	Similar study using FMG for recognition
FMG-based Study [10]	FMG-based		including sign language, finger movements, and grasps			Uses 16 FSRs; covers forearm and wrist; 10 data collections per 40 min session, session repeated
Our study (inter-trial)	nFMG-based	88.93%	Wave In, Wave Out, Pinch, Fist, Spread Fingers	12	6 s	LDA model, post-processing of 10 predictions made per second
Our study (inter-session)	pFMG-based	92.2%	Wave In, Wave Out, Pinch, Fist, Spread Fingers	12	12 s	Armband not adjusted between sessions.

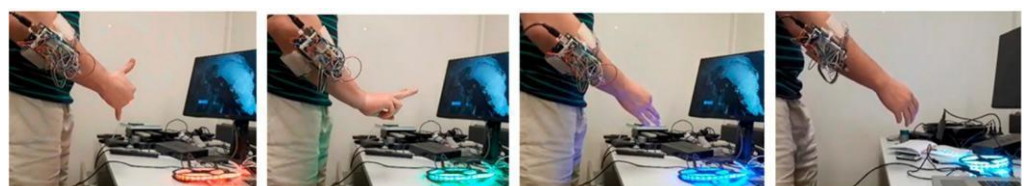


Figure 9. Demonstration of customised gestures for device teleoperation—gestures for shaka, peace, okay and rest (control) are depicted here.

3.7. Experimental Results—Online IR Device Teleoperation

The online IR device teleoperation results were limited only by the gesture recognition accuracy of the classifier and by a participant's ability to correctly point at the activating device. The prototype armband incorporated the IR transmitter onto the top of the armband placed on the forearm and thus the user was required to physically point their arm whilst activating a gesture. This partially limited gestures to the longitudinal axis. A real application may benefit from using IR boosters/extenders to allow more freedom of rotational movement and/or incorporating a more powerful IR transmitter with increased dispersive capabilities.

A variety of devices were able to be activated throughout the duration of this study, demonstrating an ability to emulate a range of functionalities. These devices (shown in Figure 10) included televisions, media systems, temperature units, entertainment/gaming systems and utilised manufacturer code types. Although incorporating compatibilities for 17 unique protocol types, it was observed that the majority of devices fell under only a few types, the primary one being the NEC protocol.

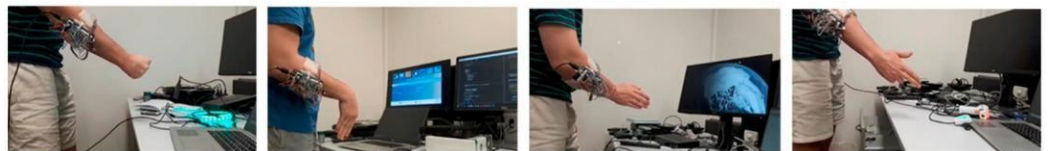


Figure 10. Operation of an LED strip, TV set box, movie hard drive player and laser tag equipment, from left to right. Also demonstrates some more custom gestures that are intuitive to the application (e.g., pistol gesture for shooting mechanic in laser tag game).

3.8. Experimental Results—Extended Framework

Spatial augmentation of the system was deemed to be sufficient for the short-term goal of attaining quasi-dynamic gestures. Users were able to perform a static gesture and then extend the functionality of that gesture by modifying their arm orientation. As demonstrated in Figure 11, a 'Fist' gesture was able to turn an LED strip on and, with longitudinal orientation of the arm, modify the LED colours during the post-prediction phase. Lateral and medial rotation of the arm changed the LED strip to red and blue, respectively. Figure 11 also demonstrates how this augmented functionality is a post-prediction process, that is, if a gesture is not identified the device is not "locked on" and therefore orientation does nothing.

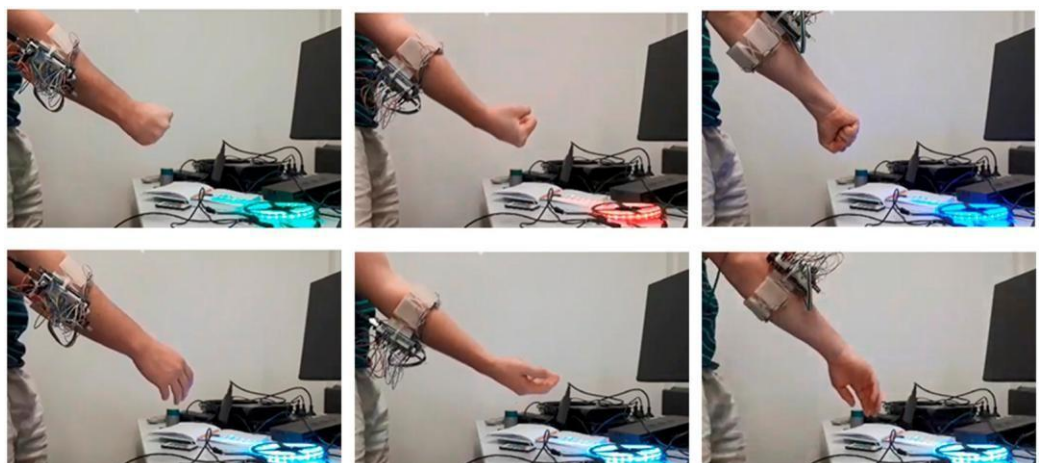


Figure 11. Spatial augmentation demonstration—**Top row:** Fist gesture augmented by arm orientation changes LED light colour. **Bottom row:** Without activating a gesture, the orientation does nothing.

Figure 11 showcases a discrete implementation of the spatial augmentation capabilities, that is, the LED strips can be in the colour state of green, red or blue with no intermediate

colours between states. A continuous gradient of colour change was deemed inefficient using quasi-dynamic gestures alone. The implemented proportional control system allowed for a more continuous variable that was proportionate to the difference between the orientated gesture position and the initial gesture position. This variable produced a spectrum of possible boundaries, the resolution determined by the user's desire. A DVD player's fast-forwarding functionalities were tested using proportional control. The more a 'Wave Out' gesture was laterally rotated, the faster the fast-forwarding function would operate. An issue that arose from this, however, was that the user's hand may initially be facing the device, but varying the orientation affected the IR transmitter's line of sight to the device. Whilst IR communication can generally be achieved across a wide field of view at short distances, it may be preferable to limit quasi-dynamic gestures to rolling motions (as opposed to pitch or yaw). Similar to the original framework, possible solutions beyond the prototype armband used would be to incorporate IR extenders/boosters, move the IR transmitter to somewhere more practical on/in the body/environment or utilise a more powerful IR transmitter with a greater field of view.

3.9. Final Remarks on Experimental Results

The key point to note from the results gathered and its subsequent comparison with existing results in the literature is that our designed framework either meets or exceeds the recognition capabilities deemed acceptable in the literature. Our armband uses fewer sensors (five) than most of the frameworks analysed in the literature. The number of gestures tested in this study was sometimes significantly lower than the research work that was designed for sign language capabilities. As no more than five gestures were tested at a time, it is presently unknown what our gesture recognition scores would be for a sign language application. For our system, it is deemed sufficient that the most popular remote-control functionalities can be served by simply a handful of gestures. Those works that incorporated more gestures, however, also tended to have significantly larger datasets for training and testing. One of the main features of our system is the turnaround time between users picking up the armband for the first time and ultimately controlling all their household devices. This can be carried out effectively within a matter of minutes—this time can be shortened even further if the user has previously saved their gesture data.

It should also be noted that the number and selection of gestures were limited to allow for easy comparison to EMG-based gesture recognition systems. Lessons learned from the testing and results of the original framework highlighted a need for better-defined static gestures. Traditionally, FMG has been compared to EMG due to the similarity in data acquisition (reading muscular signals) and its potential applications. As a result, gesture recognition methods and evaluation parameters employed for FMG have historically been guided by the standards tailored to EMG applications. It is recognised that FMG presents a substantially different signal profile to EMG, and thus it would be beneficial to explore gestures that are uniquely beneficial to FMG applications.

4. Conclusions

Both IR remote emulation and FMG-based gesture recognition systems were possible and existed in the industry/literature. What was lacking, however, was a demonstration of the amalgamation of these two domains. To the best of the authors' knowledge, no other studies have detailed or implemented an FMG-based gesture recognition system for the teleoperation of consumer IR devices. This study has described and built the foundations of designing, developing and implementing such a system as a safe and wearable human-machine interface.

This study has employed a novel pFMG armband to acquire FMG data from the upper forearm of users. This, in combination with off-the-shelf components for infrared emulation, has established a low-cost, customisable and effective acquisition device for user intent and device communication. Each of the phases of the system architecture is detailed and an in-depth analysis is provided as to the reasoning and justification for each

process followed throughout the duration of this work. A particular emphasis was placed on creating a solution that was modular, adapted to the individual user and sufficiently fast to allow for ‘plug and play’ capabilities.

Two implementations of the gesture recognition system were described in this study—the original framework and the extended framework. The original framework used five static gestures that were prevalent in myography-based gesture recognition to demonstrate the effectiveness of our proposed method. The extended framework provided additional functionalities to the gestures by spatially augmenting the data to create quasi-dynamic gestures and allow for proportional control. The efficacy and performance of the frameworks were evaluated by the pFMG data obtained from 12 and 3 participant subjects for the original and extended frameworks, respectively.

In summary of the results, the optimal features for use in the gesture recognition system were found to be RMS, IAV, SSI, VAR, MAV1, MAV2 and MAV3. The optimal classifier model for both inter-trial and inter-session accuracy was found to be the LDA model. For a dataset containing 6 s worth of pFMG data per gesture, our recognition system was able to achieve an inter-trial accuracy score of $88.93 \pm 8.70\%$. For a dataset containing 12 s worth of pFMG data per gesture, our recognition system was able to achieve an inter-session accuracy score of $92.20 \pm 9.35\%$. Teleoperation of consumer IR devices was demonstrated through successful control of a variety of consumer devices, utilising a range of protocol modulation methods and manufacturer code types. All gestures used in the original framework were able to be augmented with spatial data and with proportional control, providing extended capabilities when teleoperating consumer IR devices.

When compared to the existing literature incorporating FMG- or EMG-based gesture recognition systems with similar gesture classes, our work has proven itself to meet or even exceed results with respect to recognition accuracy and prediction speed. It should be noted that even better accuracy scores are achieved when incorporating post-processing functionalities. Aside from recognition accuracies, the implemented system has the benefit of being very customisable, affordable, lightweight, 3D printed, impervious to sweat and does not require direct skin contact.

Author Contributions: Conceptualization, G.A.; methodology, S.Y. and H.Z.; software, S.Y.; validation, S.Y.; formal analysis, S.Y.; investigation, S.Y.; resources, G.A.; data curation, S.Y. and G.A.; writing—original draft preparation, S.Y.; writing—review and editing, H.Z. and G.A.; visualization, S.Y.; supervision, H.Z. and G.A.; project administration, S.Y.; funding acquisition, G.A. All authors have read and agreed to the published version of the manuscript.

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Informed Consent Statement: Written informed consent has been obtained from the patient(s) to publish this paper.

Data Availability Statement: The data is unavailable due to privacy or ethical restrictions.

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ANNEXURE II
PROJECT ACHIEVEMENTS



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ANNEXURE III PLAGIARISM REPORT

